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**MATHEMATICAL MODELING AND ANALYSIS OF
BREAST CANCER–IMMUNE DYNAMICS WITH
CHEMOTHERAPY TREATMENT**

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Declaration

I, Tariro Kazingizi, confirm that this project report faithfully presents the results of my independent research conducted during my final year at the University of Zimbabwe. This document is submitted in partial fulfilment of the requirements for the award of the Bachelor of Science Honours Degree in Mathematics and Computational Science. I further declare that the work presented in this report is my own original work. All sources of information and ideas obtained from other authors have been properly acknowledged and cited.

Student's Signature

Date

Dedication

This thesis is dedicated to the loving memory of those who lost their lives to breast cancer. Their courage and strength in the face of such a difficult disease will always be remembered, and their memory continues to inspire the pursuit of knowledge and hope for better understanding and treatment of this illness. I also dedicate this work to my family, whose unwavering support, encouragement, and belief in my abilities have given me the strength to pursue and complete this academic journey. Their love, patience, and constant motivation have played a significant role in making this achievement possible.

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Abstract

Breast cancer remains one of the leading causes of cancer-related mortality worldwide, highlighting the need for improved understanding of tumor growth and treatment dynamics. This study develops and analyzes a mathematical model describing the interactions between breast cancer cells, immune responses, and chemotherapy treatment. The model is formulated as a system of ordinary differential equations representing tumor cells, cytotoxic T-lymphocytes, natural killer cells, stress-activated $\gamma\delta$ T cells, and the effects of chemotherapy.

Analytical techniques were used to investigate the equilibrium points and stability properties of the system, including the tumor-free equilibrium and tumor-immune coexistence states. Numerical simulations were conducted to examine the impact of chemotherapy intensity and timing of detection on tumor progression. The results show that early detection significantly improves treatment outcomes by reducing the initial tumor burden and allowing the immune system to assist in tumor suppression.

The simulations also demonstrate that adaptive chemotherapy strategies can effectively control tumor growth while reducing cumulative drug exposure compared to constant chemotherapy treatments. Overall, the findings highlight the importance of integrating immune dynamics and treatment strategies in mathematical models to better understand breast cancer progression and support improved therapeutic approaches.

Nomenclature

Symbol	Description
$x(t)$	Tumor cell population at time t
$y(t)$	Cytotoxic T-Lymphocyte (CTL) population
$N(t)$	Natural Killer (NK) cell population
$G(t)$	Stress-activated $\gamma\delta$ T cell population
$D(t)$	Chemotherapy drug concentration in the body
$u(t)$	Chemotherapy administration rate (treatment control)
r	Intrinsic tumor growth rate
γ_1	Tumor killing rate by CTL cells
γ_2	Tumor killing rate by NK cells
γ_3	Tumor killing rate by $\gamma\delta$ T cells
μ_y	Natural death rate of CTL cells
μ_N	Natural death rate of NK cells
μ_G	Natural death rate of $\gamma\delta$ T cells
κ_x	Chemotherapy-induced death rate of tumor cells
κ_y	Chemotherapy-induced suppression of CTL cells
κ_N	Chemotherapy-induced suppression of NK cells
κ_G	Chemotherapy-induced suppression of $\gamma\delta$ T cells
$\phi(x, D)$	Stress activation function for $\gamma\delta$ T cells
E^*	Tumor-immune coexistence equilibrium
TFE	Tumor-Free Equilibrium

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Chapter 1

Introduction

Breast cancer represents a major global health challenge, affecting women across all ages after puberty, with incidence rates increasing late in life. In recent years, global figures have consistently shown a high burden: estimates for 2022 indicated approximately 2.3 million new diagnoses and 670,000 deaths worldwide, paralleled by similar figures for 2021 (2.12 million cases and 674,199 deaths), (World Health Organisation, 2025 and (Deng et al. 2025). While the period between 1990 and 2021 saw an increase in global incidence and prevalence, overall mortality and disability-adjusted life years (DALYs) have thankfully declined, though significant variations exist across different regions, nations, age groups, and genders, (Deng et al. 2025). Strikingly, the data reveals significant socioeconomic inequities: countries with a very high Human Development Index (HDI) report a much higher lifetime diagnosis risk (1 in 12 women) but a lower lifetime death risk (1 in 71 women). Conversely, low HDI countries face the grim reality of a comparatively lower diagnosis rate (1 in 27 women) but a substantially higher death rate (1 in 48 women) from the disease, highlighting critical global disparities in prevention, detection, and treatment outcomes, (Kim et al. 2025).

While female gender is the most significant risk factor, accounting for approximately 99% of all cases (with men making up the remaining 0.5–1%), a variety of factors contribute to breast cancer risk, (Kim et al. 2025). These risk elements include increasing age, obesity, harmful alcohol use, a family history of the disease, prior radiation exposure, tobacco use, and postmenopausal hormone therapy. Reproductive history, specifically the age menstruation began and the age of first pregnancy, also plays a role, (Kim et al. 2025). Notably, about half of breast cancers develop in women over 40 who possess no other identifiable risk factors beyond their sex and age, (Kim et al. 2025). Although a family history elevates risk, the majority of women diagnosed do not have one; conversely, lacking a known family history does not guarantee a reduced risk. The management and treatment principles for breast cancer in men are essentially the same as those used for

women, ([Kim et al. 2025](#)).

Breast cancer develops when abnormal cells in the breast begin to grow and divide uncontrollably. Under normal conditions, healthy cells divide to repair or replace damaged tissue and stop dividing or die when appropriate. However, cancerous breast cells ignore these regulatory signals, continuing to multiply and forming a mass called a tumor. Over time, these cancerous tumors can spread to other areas of the body, ([Breastcancer.org, 2025](#)).

Understanding the breast cancer micro environment and the interactions between cancer cells and their treatments is essential for improving patient outcomes. Mathematical modeling has emerged as a valuable approach for studying the dynamics of key cells and cytokines involved in breast cancer progression. This study seeks to use existing mathematical models to estimate optimal treatment strategies, assess the impact of early detection and chemotherapy, and provide insights into tumor behavior that may not be easily obtained through laboratory experiments alone.

1.1 Development of breast cancer cells

Breast cancer begins when genetic mutations occur in the DNA of breast cells, disrupting normal cell growth and division processes. Under normal conditions, breast cells grow and divide in an orderly manner, replacing old or damaged cells as needed. However, mutations in specific genes—such as BRCA1, BRCA2, TP53 and the amplification of the HER2 gene can alter the regulation of cell growth, leading to uncontrolled proliferation and the distinct biological subtypes observed clinically [Yersal & Barutca \(2014\)](#).

These abnormal cells do not respond to signals that regulate cell death (apoptosis) or limit cell division. As a result, they continue to multiply, forming a cluster or tumor. As genetic damage accumulates, the malignant cells acquire capabilities such as sustained proliferative signaling and evasion of growth suppressors [Hanahan & Weinberg \(2011\)](#), gaining the ability to invade nearby tissues and spread (metastasize) via the bloodstream or lymphatic system. The dynamic evolution of this cell population—the balance between cell proliferation and elimination is precisely the process that must be understood and quantified when assessing the efficacy of treatment [Turner et al. \(2020\)](#).

The development of breast cancer is a multistep process influenced by genetic, molecular, and environmental factors. Hormonal influences particularly estrogen and progesterone can stimulate cell division in breast tissue, increasing the chance of accumulating mutations over time (Russo, 2006). Other risk factors such as age, family history, lifestyle, and exposure to carcinogens also contribute to these changes.

1.1.1 Types of Breast Cancer

Breast cancer begins when DNA changes (mutations) occur in normal breast cells—most commonly those lining the milk ducts (ductal carcinoma) or the milk-producing lobules (lobular carcinoma). These mutations cause the cells to grow and divide uncontrollably, forming an abnormal mass of tissue known as a tumor. Initially, the cancer cells may remain confined to their original location (in situ), but if the cancer is invasive, these cells break out and infiltrate the surrounding breast tissue. Over time, malignant cells can detach from the primary tumor and enter the body's lymphatic system or bloodstream. Once in these vessels, the cancer cells can travel to nearby lymph nodes, most often those in the underarm (axilla), and eventually spread to distant organs like the bones, lungs, liver, or brain, a process called metastasis [Standing \(2021\)](#)

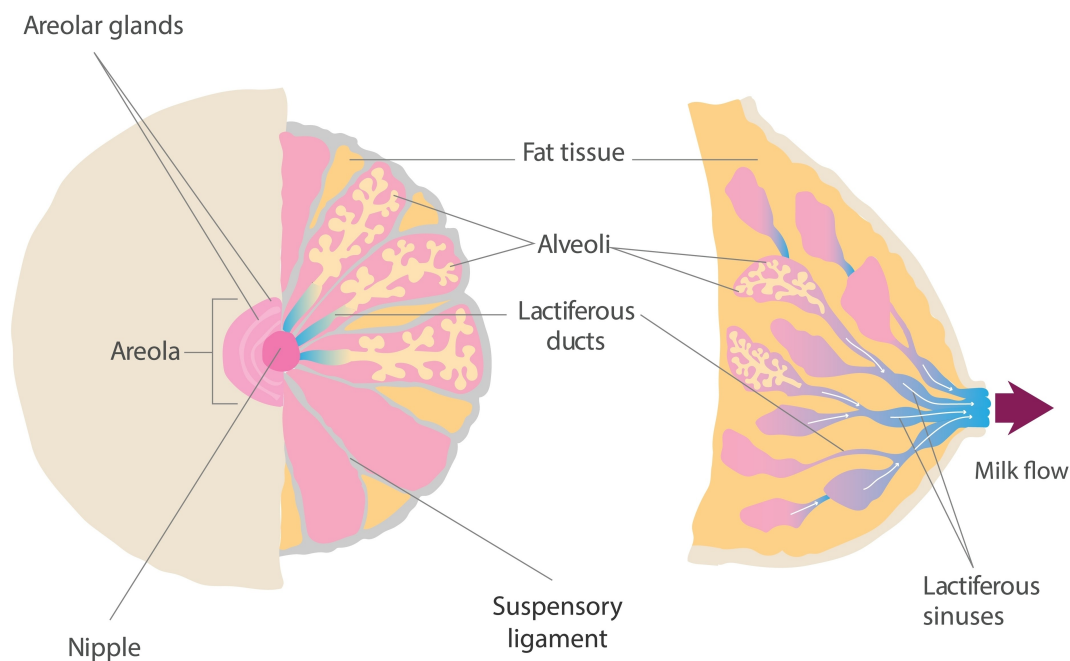


Figure 1.1: Structure of the Human Breast: The illustration depicts the mammary glands, including the network of ducts and the lobules. Breast cancer typically begins in the epithelial lining of these ducts or lobules before potentially spreading to surrounding tissues or the lymph nodes.

[Standing \(2021\)](#)

Below are the main types of breast cancer:

1. Ductal Carcinoma In Situ (DCIS)

Ductal Carcinoma In Situ (DCIS) represents the non-invasive or pre-invasive end of the breast cancer spectrum, often formally classified as Stage 0 disease. Pathologi-

cally, DCIS is defined by the unrestricted proliferation of malignant epithelial cells that remain strictly confined within the ductal system, crucially maintaining the integrity of the basement membrane. Because the malignant cells lack access to the vascular and lymphatic systems, this condition is non-metastatic and not immediately life-threatening. Clinically, DCIS rarely presents as a palpable mass, with the overwhelming majority of cases detected through routine screening mammography, where it is often visualized as clusters of pleomorphic or punctate microcalcifications. Given its classification as an obligate precursor, the primary therapeutic goal is to eliminate the lesion to mitigate the significant risk of subsequent progression to invasive disease, typically involving surgical excision with or without adjuvant radiation therapy. [National Breast Cancer Foundation \(NBCF\) \(2025\)](#).

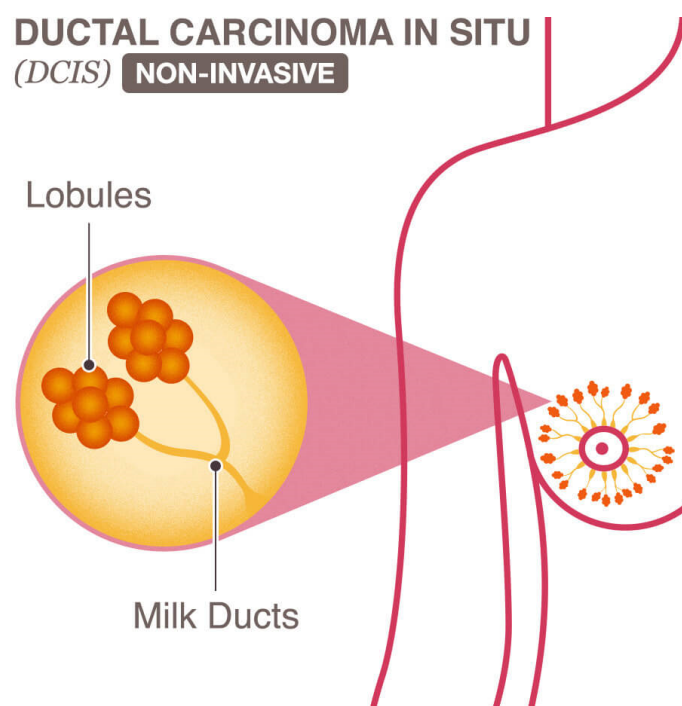


Figure 1.2: Ductal Carcinoma In Situ (DCIS): Non-invasive cancer confined to the ducts.

2. Invasive Ductal Carcinoma (IDC)

Invasive Ductal Carcinoma (IDC), also frequently referred to as Infiltrating Ductal Carcinoma, constitutes the most common form of invasive breast cancer, representing approximately 70% to 80% of all invasive diagnoses. The critical feature that distinguishes IDC is the definitive breach of the ductal basement membrane by the malignant cells, allowing them to infiltrate the surrounding mammary stroma. This structural invasion is clinically paramount because it grants the cancer cells access to the lymphatic and circulatory networks, thereby establishing the potential for both regional metastasis to the lymph nodes and distant metastasis to visceral organs. Due to this metastatic capacity, IDC is staged from I to IV, and its manage-

ment is necessarily aggressive, encompassing surgery (lumpectomy or mastectomy) followed by complex adjuvant and/or neoadjuvant systemic therapies, including chemotherapy, targeted agents, and endocrine therapy, which are tailored to the specific molecular characteristics of the tumor [Cleveland Clinic \(2024\)](#).

3. Invasive Lobular Carcinoma (ILC)

Invasive Lobular Carcinoma (ILC) is recognized as the second most common type of invasive breast cancer, accounting for approximately 10% to 15% of all invasive diagnoses. Histologically, ILC originates in the lobules, which are the milk-producing glands of the breast. The "invasive" designation signifies that the malignant cells have breached the basement membrane of the lobule and extended into the surrounding mammary stroma (fibrous and fatty tissue). This invasion typically presents a distinct challenge in diagnosis because ILC cells often lose the expression of the adhesion protein E-cadherin, preventing them from clustering into a cohesive mass. Instead, the cells characteristically infiltrate the breast tissue in a single-file, linear growth pattern or in sheets, leading to an insidious presentation. This diffuse growth means that ILC is less likely to manifest as a firm, palpable lump, often presenting instead as a subtle area of breast thickening, fullness, or architectural distortion that can be difficult to detect during physical examination or standard mammographic screening. Therefore, advanced imaging modalities, such as breast MRI, are frequently required to accurately determine the true extent of the disease and ensure appropriate staging and management [City of Hope/Cancer Treatment Centers of America \(2024\)](#).

1.1.2 Subtypes of Breast Cancer

The classification of breast cancer into molecular subtypes is crucial for prognosis and guiding therapeutic strategies. This classification is primarily based on the immunohistochemical (IHC) expression of three key receptors: Estrogen Receptor (ER), Progesterone Receptor (PR), and Human Epidermal Growth Factor Receptor 2 (HER2)

1. Luminal A

Luminal A tumors are the most common subtype and are characterized by being Estrogen Receptor-positive (ER+), Progesterone Receptor-positive (PR+), and HER2-negative (HER2-). They are also typically associated with a low Ki-67 proliferation index (a marker for how quickly cancer cells are dividing). This profile suggests that the cancer growth is highly driven by hormones, making the tumors generally slower-growing and less aggressive compared to other subtypes. Because

of their strong hormone receptor positivity, the standard and most effective systemic treatment for Luminal A breast cancer is endocrine (hormone) therapy, often in combination with surgery and radiation, and sometimes chemotherapy depending on other risk factors. This subtype is associated with the best overall prognosis [Dai et al. \(2020\)](#)

2. Luminal B

Luminal B tumors share the characteristic of being Hormone Receptor-positive (HR+)—meaning they are ER+ and/or PR+—but they differ from Luminal A in having either high expression of the Ki-67 proliferation index or by being HER2-positive (HER2+). In practice, Luminal B is often subcategorized into Luminal B HER2-negative (HR+, HER2-, high Ki-67) and Luminal B HER2-positive (HR+, HER2+, any Ki-67). This subtype is generally more aggressive and faster-growing than Luminal A. Treatment is more intensive and involves endocrine therapy along with chemotherapy (due to the higher proliferative rate) and, if HER2-positive, HER2-targeted therapy. The prognosis for Luminal B is generally less favorable than Luminal A, reflecting the higher proliferative activity and potential for HER2-driven growth [City of Hope/Cancer Treatment Centers of America \(2024\)](#).

3. HER2-Enriched

The HER2-Enriched subtype is defined by being HER2-positive (HER2+) and Hormone Receptor-negative (ER- and PR-). The over-expression or amplification of the HER2 gene leads to an excessive amount of the HER2 protein on the surface of the cancer cells, which aggressively promotes cell growth and division. These tumors are considered highly aggressive and fast-growing. Historically, this subtype had a poor prognosis, but the introduction of HER2-targeted therapies (such as trastuzumab and pertuzumab), often combined with chemotherapy, has dramatically improved outcomes, making it one of the most treatable subtypes [Dai et al. \(2020\)](#).

4. Triple-Negative Breast Cancer (TNBC)

Triple-Negative Breast Cancer (TNBC) is characterized by the absence of all three receptors: it is Estrogen Receptor-negative (ER-), Progesterone Receptor-negative (PR-), and HER2-negative (HER2-). Lacking these receptors means that the cancer cells do not respond to targeted therapies like endocrine therapy or HER2-targeted drugs. TNBC is often associated with a basal-like molecular profile, high histological grade, and is more common in younger women and those with a BRCA1 gene mutation. It is typically the most aggressive subtype, with a higher risk of recurrence, especially early on. Treatment relies primarily on chemotherapy, surgery, and radiation, with newer options including immunotherapy and PARP inhibitors

for specific patient subgroupCity of Hope/Cancer Treatment Centers of America (2024).

1.1.3 Treatment of Breast Cancer

Breast cancer treatment is highly individualized, combining both **local therapies** (treating the breast and nearby lymph nodes) and **systemic therapies** (treating the entire body)(NCCN). The following are the therapies for breast cancer;

Table 1.1: Overview of Breast Cancer Treatment Modalities

Therapy Type	Description and Mechanisms
Surgery (Local Therapy)	Lumpectomy: Removes only the tumor and a margin of healthy tissue. Preserves most of the breast. Almost always followed by radiation therapy. Mastectomy: Removes the entire breast. Recommended for larger/multiple tumors or <i>BRCA</i> mutations. Lymph Node Surgery: Sentinel Biopsy (preferred) or Axillary Dissection to check for cancer spread while minimizing lymphedema risk.
Radiation Therapy (Local Therapy)	Uses high-energy X-rays to kill remaining cells. External Beam Radiation: Most common; delivered to breast or chest wall over several weeks. Internal Radiation (Brachytherapy): Radioactive sources placed inside the breast cavity to shorten treatment time.
Chemotherapy (Systemic Therapy)	Uses cytotoxic drugs to target rapidly dividing cells. Non-specific, causing systemic side effects (hair loss, nausea). Standard for TNBC and HER2-Positive cancer. Given neoadjuvantly or adjuvantly.
Hormone Therapy (Systemic Therapy)	Used for ER+ and/or PR+ tumors. Blocks estrogen effects or reduces production. Includes Selective Estrogen Receptor Modulators (SERMs) like Tamoxifen or Aromatase Inhibitors (AIs) like Anastrozole. Taken orally for 5–10 years.

Continued on next page

Therapy Type	Description and Mechanisms (Continued)
Targeted Therapy (Systemic Therapy)	Precision treatment blocking specific proteins. HER2-Targeted: Essential for HER2+ tumors (e.g., Trastuzumab). PARP Inhibitors: For <i>BRCA1/2</i> mutations; blocks DNA repair. CDK4/6 Inhibitors: Used for HR+/HER2- to block division signals.
Immunotherapy (Systemic Therapy)	Boosts the immune system. Mechanism: Checkpoint Inhibitors (e.g., Pembrolizumab) release "brakes" on immune cells. Use: Primarily for advanced/aggressive cancers like TNBC, often combined with chemotherapy.

1.2 Statement of the Problem

Breast cancer remains one of the leading causes of cancer-related deaths worldwide despite significant improvements in diagnosis and treatment. The progression of breast cancer is influenced by complex interactions between tumor cells and the immune system. Immune cells such as Cytotoxic T Lymphocytes (CTLs) and Natural Killer (NK) cells play an important role in identifying and destroying cancer cells. However, treatments such as chemotherapy, while effective in reducing tumor growth, can also suppress these important immune responses, creating challenges in achieving effective and balanced treatment outcomes (De Pillis & Radunskaya 2003).

Many existing mathematical models of cancer treatment focus on simplified tumor-immune interactions or assume constant chemotherapy administration. Such assumptions may fail to capture the dynamic and nonlinear nature of tumor growth, immune response, and treatment effects (Kuznetsov et al. 1994). In addition, limited attention has been given to models that incorporate multiple immune components together with time-dependent chemotherapy dynamics.

Therefore, there is a need to develop a mathematical framework that better represents these biological interactions. This study develops a five-dimensional system of ordinary differential equations to model the interactions between breast cancer cells, immune responses, and chemotherapy treatment. Using stability analysis and numerical simulations, the study investigates how immune strength, treatment timing, and chemotherapy administration influence tumor progression, tumor control, and possible relapse dynamics.

1.3 Research Aim and Objectives

1.3.1 Aim

The primary aim of this study is to formulate and analyze a five-dimensional system of Ordinary Differential Equations (ODEs) to evaluate the synergistic impact of early detection and chemotherapy on breast cancer-immune dynamics. Specifically, the research seeks to quantify how the timing of diagnostic awareness and the rate of drug administration can optimize therapeutic efficacy while minimizing systemic toxicity and preserving host immune integrity.

1.3.2 Objectives

The objectives of this study are:-

1. To formulate an expanded five-compartment mathematical model using Ordinary Differential Equations (ODEs) that integrates the non-linear interactions between breast cancer cells, Cytotoxic T Lymphocytes (CTLs), Natural Killer (NK) cells, stress-responsive $\gamma\delta$ T cells, and chemotherapy dosage.
2. To perform a rigorous mathematical analysis of the ODE model, including the proof of positivity and boundedness of solutions, and the identification of stability conditions for the Tumor-Free Equilibrium (TFE) and the Tumor-Persistent Equilibrium (TPE) states.
3. To conduct a sensitivity analysis to identify the most influential parameters, such as the drug dosage rate (u), the immune recruitment rates (ρ), and the chemotherapy-induced killing rates (κ), that govern the transition from malignancy to recovery.
4. To perform numerical simulations using standard integration techniques (such as the Runge-Kutta method) to compare late-stage diagnosis requiring high-dose chemotherapy against early-stage detection with optimized treatment, demonstrating how early intervention reduces the systemic "side effects" and toxic burden on the host immune integrity.

1.4 Justification Of Study

Breast cancer remains one of the leading causes of cancer-related mortality worldwide, highlighting the need for improved tools to evaluate the timing and effectiveness of treatment strategies. Early detection is known to significantly improve patient outcomes

by identifying tumors at smaller and more manageable stages. However, aggressive chemotherapy can also suppress important components of the immune system such as Cytotoxic T Lymphocytes (CTLs) and Natural Killer (NK) cells that contribute to tumor elimination (De Pillis & Radunskaya 2003). Understanding the balance between tumor suppression and immune preservation is therefore essential in cancer treatment.

This study is justified by the development of a five-dimensional deterministic system of ordinary differential equations that models the interactions between tumor cells and multiple immune populations, including CTLs, Natural Killer cells, and $\gamma\delta$ T cells. Mathematical models have been widely used to study tumor-immune interactions and to identify conditions under which tumors can be controlled or eliminated (Kuznetsov et al. 1994). The proposed ODE framework provides a rate-based approach for analysing how tumor growth, immune response, and chemotherapy administration interact over time.

Using this framework, the model serves as a computational environment to explore how early detection, represented by smaller initial tumor populations, can improve treatment outcomes and reduce the need for aggressive chemotherapy. The stability analysis and numerical simulations conducted in this study therefore provide useful insights into how treatment strategies may be optimized to control tumor growth while preserving immune system function.

1.5 Chapter Overview

Chapter 1 introduces breast cancer as a major global health crisis, citing high statistics and significant socioeconomic disparities in outcomes worldwide. The female gender is the strongest risk factor (99% of cases), with other key factors including age, obesity, family history, and lifestyle. The disease develops through genetic mutations in breast cells, leading to uncontrolled proliferation and tumor formation. The chapter classifies the disease into types like Ductal Carcinoma In Situ (DCIS) and Invasive Ductal/Lobular Carcinoma, and four molecular subtypes. Current management involves individualized local and systemic therapies, including surgery, radiation, chemotherapy, and hormone/targeted agents. The core problem identified is that existing mathematical models for tumor-immune dynamics fail to integrate the crucial impact of chemotherapy dosage. This gap justifies the study's aim to develop a unified model incorporating early detection and chemotherapy dynamics.

Chapter 2

Literature Review: Mathematical Modeling of Breast Cancer

2.1 Introduction to Literature Review

Mathematical oncology is a vital new field that combines math and cancer biology. It gives us strong, numerical tools to understand the complicated, ever-changing ways cancer starts, grows, and reacts to treatment. Breast cancer is particularly difficult to treat because its cells are genetically different (heterogeneous), they constantly interact with their surroundings (microenvironment), and they quickly learn to resist drugs. These complexities overwhelm traditional clinical methods. The main purpose of this chapter is to look closely at all the mathematical models that have been created to measure these exact dynamics in breast cancer and evaluate how well different treatment plans actually work.

2.1.1 Review on Ordinary Differential Equations (ODEs) Models

Traditional Pharmacokinetic/Pharmacodynamic (PK/PD) models for new cancer drugs are limited because they treat all patients the same, making them poor at predicting individual responses, especially with small amounts of early trial data. [Xiang et al. \(2024\)](#) addressed this by developing a new method called Data-Driven Neural Ordinary Differential Equation (DN-ODE), combining machine learning with ODEs to create a highly accurate, personalized prediction model for each breast cancer patient. This smart program uses limited early data to forecast the rate of tumor shrinkage and Progression-Free Survival (PFS), achieving high confidence scores (R^2 over 0.9) in later trial predictions. Furthermore, a Principal Component Analysis (PCA) visualization confirmed the model's ability to group patients with similar tumor growth patterns, effectively identifying indi-

viduals most likely to benefit from the treatment.

A research by [Alharbi et al. \(2023\)](#) proposed a fat-estrogen-breast cancer mathematical model to investigate the combined effects of body mass, estrogen levels, and the immune system on the evolution of cancer cells. To validate the model, the researchers performed a stability analysis at various biologically plausible equilibrium states.

They reported that a tumor-free state can be sustained only if estrogen levels are normal and if the immune response rate against the tumor outweighs the tumor's growth rate in the presence of fat cells; otherwise, the tumor-free state is unstable, suggesting growth. Conversely, a co-existing equilibrium (meaning the tumor and other factors are stable together) exists when estrogen levels are elevated, provided certain mathematical conditions are met. Ultimately, the researchers concluded that whether the tumor cells are eliminated or continue to proliferate is directly dependent on the interplay between the rate of fat cell growth, the strength of the immune response, and the patient's estrogen level. These theoretical findings were then confirmed through numerical simulations.

The study by [Azizan et al. \(2024\)](#) stated that understanding the early stages of cancer progression is important because it allows women to make timely treatment choices. They focused on Cytotoxic T Lymphocytes (CTLs), which are believed to act as the immune system's primary effector cells in eliminating cancer. The relationship between CTLs and tumor cells was described as a "predator-prey" dynamic, with the CTLs acting as the predator and the tumor cells as the prey. For their mathematical model, the study involved performing a stability analysis using the Jacobian matrix on the model's steady-state solutions. This analysis was intended to map out the stability region of breast cancer cells across two different phases of the cell cycle (interphase and metaphase) to better describe tumor growth patterns in different people.

To examine the actual population development, the team applied the Fourth Order Runge-Kutta (RK4) method, a numerical technique, to estimate the populations of both the tumor cells and the CTLs over time. Their results confirmed that individuals with a higher immune system population (more CTLs) were more likely to fight off the cancer for a longer period compared to those with a poor immune system. They specifically noted that when the CTL population was high, the cancer cells were reduced in a shorter timeframe.

The authors [Saad et al. \(2025\)](#) explained that they had developed a nonlinear tumor-immune interaction model based on two coupled Ordinary Differential Equations (ODEs), which track tumor growth and the resulting immune response over time. They incorporated early detection into the model by using a saturating function that depends on both the tumor's size and the level of public awareness. The system was simplified through nondimensionalization to make the analysis easier. Following a mathematical analysis, they

reported identifying two biologically meaningful equilibrium states and established the specific conditions required for a stable tumor-free state.

Their simulations showed that increasing public awareness significantly enhanced early detection, which subsequently suppressed tumor growth. Furthermore, a sensitivity analysis confirmed that the parameters related to awareness and detection were the most crucial for controlling the tumor. The study concluded that this framework successfully quantifies how public awareness drives early detection to shape the tumor-immune outcome, emphasizing the importance of screening initiatives and public campaigns for effective cancer management.

Investigating the development of breast tumor [Owanga \(2023\)](#) announced that they were developing an ODE model for HER2-positive breast cancer that simplified both the tumor structure (by accounting for the critical interface zone where the tumor meets healthy tissue) and its interaction with the immune system. They stated that they specifically modeled the relationship between the tumor cells and Natural Killer (NK) cells, noting that the antibody drug trastuzumab can influence this interaction.

The study used the model to investigate tumor growth under two scenarios: in isolation (no treatment) and in the presence of NK cells and trastuzumab. Comparing the model's predictions to empirical data, the researchers found that simply increasing the cytotoxicity of NK cells was likely insufficient to control highly aggressive, fast-growing tumors. However, they drew the significant conclusion that if the surrounding healthy tissue maintains high levels of patrolling cytotoxic NK cells, the tumor would be unable to establish itself.

[Yakubu et al. \(2023\)](#) stated that many mathematical models developed to find the causes and cures for cancer have resulted in systems of nonlinear stiff ordinary differential equations. Using these models, they explained that they had obtained numerical estimates of biological parameters, interpreted various phenomena, and made predictions to gain further knowledge about cancer development and treatment options. In their study, they performed numerical simulations using continuous block implicit hybrid methods and reported that their results successfully supported existing theoretical and clinical findings.

By analyzing the interactions among different tumor cell populations and graphically presenting the results, they claimed to have clearly demonstrated the effects of all competing cell populations, as well as the effects of conventional chemotherapies previously used in breast cancer treatment. They acknowledged that while these past treatments played vital roles, they often incurred severe side effects in normal tissues and organs, and recent research now suggests that they often do not represent a complete cure. The study, therefore, concluded that there is a clear need to focus not only on preventative measures but also on developing more successful treatments to increase the survival rate

of breast cancer patients.

A foundational contribution to the field of mathematical oncology was made by [de Pillis et al. \(2005\)](#), who developed a comprehensive ordinary differential equation (ODE) model to characterize the complex interactions between a growing tumor and multiple branches of the immune system. Their research specifically focused on the distinct roles of Natural Killer (NK) cells and Cytotoxic T Lymphocytes (CTLs), modeling the "predator-prey" relationship where immune cells actively seek and destroy malignant "prey". By incorporating experimental data into their simulations, the authors demonstrated that while the innate immune response (NK cells) provides an immediate defense, the adaptive response (CTLs) is essential for long-term tumor suppression. This framework has become a benchmark for subsequent studies, providing the mathematical basis for understanding how different immune populations compete and cooperate to control tumor progression, as well as evaluating the efficacy of external interventions like chemotherapy.

A significant advancement in the precision of oncological modeling was introduced by [Idrees et al. \(2024\)](#), who utilized the Caputo–Fabrizio fractal-fractional derivative to study breast cancer dynamics. Their work moved beyond the limitations of classical integer-order calculus by incorporating two critical parameters: the fractional order (α), which captures the "memory effect" or hereditary traits of biological systems, and the fractal dimension (β), which accounts for the complex, non-Euclidean geometry of the tumor microenvironment. By applying this framework to a tumor-immune system, the authors demonstrated that the fractal-fractional approach provides a more realistic representation of how cancer cells proliferate and interact with host defenses over time. Their findings confirmed that the hereditary properties of cells and the jaggedness of the tumor frontier significantly influence the stability of the disease, providing a modern mathematical foundation for evaluating the long-term effectiveness of clinical interventions.

2.1.2 Review on Partial Differential Equations (PDEs)

The histopathological analysis is the standard for breast cancer diagnosis, but [Tyagi et al. \(2025\)](#) noted that the quality of microscopic images can be corrupted by issues like Poisson noise, artifacts, and poor contrast, which limit the accuracy of a pathologist's findings. To address this, they proposed an optimized Fourth Order Partial Differential Equation (PDE) filter. They explained that they enhanced the filter by using the Cuckoo Search Optimization technique to automatically set the edge detection parameters, which avoids relying on user input. The procedure involved first converting the microbiopsy images to the L*a*b* color space and applying Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance abnormalities. The proposed PDE filter was then applied to remove noise and preserve important features like cell boundaries (edges).

Finally, the restored image was merged with the other color channels and converted back to a standard format. The study reported that their method achieved significant quantitative improvements, demonstrating an average percentage improvement in a key quality metric (PSNR) of 16.24% and 23.39% on two major breast tumor image datasets (BreakHis and BreCaHAD), confirming the strategy's effectiveness in cleaning, enhancing, and restoring image information within a single framework.

The distributed parallel computing systems offer extensive computing and storage resources, [Alias et al. \(2009\)](#) provided an opportunity to develop new, dynamic solvers for Partial Differential Equations (PDEs), which are necessary for visualizing cancerous breast cells. They explained that while PDEs (which include parabolic, elliptic, and hyperbolic equations) are commonly used in many scientific fields, their work would specifically focus on elliptic equations (like Helmholtz's wave equation, which can model electromagnetic waves used in biological imaging) and hyperbolic equations. To find the numerical solutions, they chose the finite-difference method for discretizing the equations. Furthermore, the team emphasized the use of Parallel Virtual Machine (PVM) software to manage communication across the distributed system, which they anticipated would result in a performance gain. The overall goal of the study was to analyze the parallel performance measurements achieved by using these methods.

[Park & Yang \(2018\)](#) emphasized the significance of their study by highlighting the importance of nondestructive testing for detecting defects and ensuring the reliability of breast cancer prediction through thermal analysis. They acknowledged that most current nondestructive tests negatively affect the human body and lack adequate precision. To increase the reliability of these methods, the team constructed a theoretical model using two-dimensional modeling and Partial Differential Equation (PDE) thermal analysis. They explained that this model simulated inner breast tissue to allow for non-wounding observation and estimation of the presence and extent of simulated breast cancer based on the thermal patterns.

The researchers reported that their theoretical analysis confirmed the possibility and preference for using this non-wounding defect detection method. Furthermore, the main contribution of their work was identified as the ability to provide long-term estimates, noting that the model could be expanded to predict breast cancer purely through heat analysis and potentially be applied to various other cancers and tumors in the body.

The researchers Mohammad [Mirzaei et al. \(2022\)](#) explained that since the development of breast tumors heavily relies on the complex interaction network among different cell types (including immune and cancer cells), their study aimed to develop a data-driven mathematical model that uses newly collected spatio-temporal mouse data to account for the cells' physical location within the tumor. Their results showed that cancer cells had

a minor presence in the areas with the highest overall concentration of immune cells.

They observed that when there was no influx of immune cells, the number of activated immune cells inside the tumor depleted over time. Intriguingly, when the model included the influx of immune cells (using a Robin boundary condition), the highest concentrations of both T cells and cancer cells were found at the boundary of the tumor, implying that the immune cell influx actually caused a dominant outward movement (advection) of cancer cells. Finally, the team used sensitivity analyses to determine that the diffusion rates of cancer cells and adipocytes (fat cells) were the most critical parameters, followed closely by the influx and diffusion rates of cytotoxic T cells, suggesting that targeting these specific rates could be a viable strategy for treating breast cancer.

[Geweid & Abdallah \(2019\)](#) aimed to improve the accuracy of recognition in mammogram images using a novel nonparametric approach called IBCNP (Identification of Breast Cancer using a Nonparametric approach). They explained that the nonparametric approach is a new method for dealing with the uncertainty inherent in cancer detection without imaging. The study investigated the feasibility of applying this technique to distinguish between normal and malignant pixel intensities in mammography, framing the detection problem as maximizing the shared data between the image pixels and the infected region, subject to a constraint on the desired area's border size.

The team conducted a comprehensive comparative study of their proposed algorithm against four existing medical image segmentation techniques, including one based on Partial Differential Equations (PDEs), and applied all five methods to 156 different mammogram images. Evaluating the results using multiple metrics like image quality (Q), the Dice similarity coefficient (DSC), and the probability of tumor detection (PTD), the researchers concluded that the proposed IBCNP algorithm was 94.937% accurately effective in breast cancer detection.

Research conducted by [Malik Basha & Sashirekha \(2025\)](#) aimed to explain that the primary goal of their study was to assess the reliability and specificity of forecasting breast cancer using a Non-Partial Differential Equation (Non-PDE) algorithm in comparison to the well-known Perona-Malik scheme. They reported that they implemented a non-partial divergent problem on a health data collection to establish a new prediction methodology. The comparison between the two strategies on the examined sample set showed that the Perona-Malik scheme achieved a diagnostic rate of 93%, while the Non-PDE approach achieved a slightly higher prediction rate of 94%. They noted that this small difference was statistically significant and concluded that the Non-PDE algorithm performed significantly better than the influential Perona-Malik method when comparing both precision and accuracy.

According to [Saravanan & Basha \(2023\)](#), the researchers reported that the aim of their

study was to evaluate the precision and correctness of chest cancer prediction by comparing the Anisotropic Diffusion Algorithm (ADA) against a Non-Partial Differential Equation (Non-PDE) Denoising Algorithm.¹ Using a cancer dataset with a small sample size of five, they found that the Non-PDE denoising classifier produced an accuracy of 92% for predicting the disorder. In contrast, they reported that the Anisotropic Diffusion Algorithm achieved a higher prediction accuracy of 94.80%, a difference they noted was statistically significant ($2p < 0.05$).³ The study ultimately concluded that the Anisotropic Diffusion Algorithm performed better than the Non-PDE Denoising method, demonstrating higher overall accuracy.

2.2 Chapter Overview

Chapter 2 reviews mathematical oncology methods for studying breast cancer dynamics, focusing on two main approaches: Ordinary Differential Equations (ODEs) and Partial Differential Equations (PDEs). ODE models cover personalized prediction, tumor-immune system interaction (often predator-prey dynamics), and the combined effects of hormonal and lifestyle factors. Existing ODE frameworks have successfully quantified how public awareness drives early detection in controlling tumor growth. Conversely, PDE models are used for spatial analysis, including image processing to enhance micro-biopsy quality, and for non-wounding thermal analysis of tumors. Spatiotemporal PDE models detail how cell movement and diffusion rates (e.g., T cells) critically influence cancer cell dynamics. The chapter establishes the foundation for the current study, validating the use of ODE models while highlighting the need to integrate the unaddressed factor of chemotherapy dosage.

Chapter 3

Model Formulation and Analysis

3.1 Introduction

In this chapter, the study introduces a comprehensive mathematical model designed to analyze the dynamics of breast cancer, specifically examining the interactions between malignant cells, the multi-branched immune system, and systemic chemotherapy. The framework details the model's structure, including its biological assumptions and compartmental interactions, and utilizes a system of first-order ordinary differential equations (ODEs) to govern its behavior. A key feature of this model is the explicit incorporation of a constant chemotherapy infusion (u) and a specialized stress-activation function $\phi(x, D)$ that represents the recruitment of unique $\gamma\delta$ T cells. Following the formulation, the system's behavior is decoded through rigorous analyses, including stability analysis to identify critical biological thresholds for a stable, tumor-free state and sensitivity analysis to rank parameters like immune killing efficiency and drug efficacy.

The model is built upon several core biological and mathematical assumptions. Biologically, it assumes that breast cancer cells (x) proliferate logistically within the carrying capacity of the local environment. The interaction between the tumor and immune effectors—including Cytotoxic T Lymphocytes (CTLs), Natural Killer (NK) cells, and $\gamma\delta$ T cells—is modeled as a predator-prey functional response. Additionally, chemotherapy (D) is assumed to induce cellular stress that recruits $\gamma\delta$ T cells (G) while simultaneously exhibiting non-specific systemic toxicity that can exhaust healthy immune compartments. Mathematically, the model assumes a homogeneous, "well-mixed" population where all components are uniformly distributed, and it treats the chemotherapy infusion rate (u) as a constant to allow the drug concentration to reach a steady-state equilibrium for stability analysis.

3.2 Model Assumptions

The formulation of the breast cancer-immune ODE model is based on the following biological and mathematical assumptions:

3.2.1 Biological Assumptions

1. **Logistic Tumor Growth:** It is assumed that breast cancer cells (x) proliferate logistically, where growth is limited by the carrying capacity of the local tissue environment.
2. **Immune Predator-Prey Dynamics:** The interaction between the immune effectors (CTLs, NK cells, and $\gamma\delta$ T cells) and the tumor is assumed to follow a predator-prey functional response, where immune cells actively seek and destroy malignant cells.
3. **Synergistic Stress Recruitment:** A central assumption is that chemotherapy (D) induces cellular stress on the tumor, which acts as a biochemical signal to recruit stress-responsive $\gamma\delta$ T cells (G) through the function $\phi(x, D)$.
4. **Systemic Toxicity:** Chemotherapy is assumed to be non-specific, meaning it simultaneously reduces the tumor population and induces “friendly fire” decay or exhaustion in the healthy immune compartments.

3.2.2 Mathematical Assumptions

1. **Homogeneous Population:** The model assumes a “well-mixed” system where cancer cells, immune cells, and the drug are uniformly distributed throughout the breast tissue.
2. **Constant Drug Administration:** For the purpose of stability analysis, the chemotherapy infusion rate (u) is assumed to be a constant value, allowing the drug concentration (D) to reach a steady-state equilibrium.
3. **Memoryless Dynamics:** By utilizing integer-order ODEs, the model assumes that the rate of change in cell populations depends only on their current state, without considering historical memory effects.

3.3 Parameter Definition

Table 3.1: Parameter Definitions

Symbol	Definition
r	Intrinsic growth rate of cancer cells
K	Tumor carrying capacity
γ_1	Killing rate of tumor cells by CTLs (y)
γ_2	Killing rate of tumor cells by NK cells (N)
γ_3	Killing rate of tumor cells by $\gamma\delta$ T cells (G)
κ_x	Chemotherapy-induced death rate of tumor cells
s_y	Constant source rate of CTLs
s_N	Constant source rate of NK cells
s_G	Constant source rate of $\gamma\delta$ T cells
ρ_y	Maximum recruitment rate of CTLs
ρ_N	Maximum recruitment rate of NK cells
ρ_G	Stress-induced recruitment rate of $\gamma\delta$ T cells
μ_y	Natural death rate of CTLs
μ_N	Natural death rate of NK cells
μ_G	Natural death rate of $\gamma\delta$ T cells
η	CTL inactivation rate by tumor cells
m, k, a	Half-saturation constants for cell recruitment
$\kappa_y, \kappa_N, \kappa_G$	Chemotherapy-induced death rate of immune cells
u	Constant chemotherapy dose rate
λ	Drug clearance rate

3.4 The Compartment Diagram

The mathematical model is structured around a five-dimensional system that captures the essential interactions between the tumor microenvironment and clinical intervention. At the core of the system are the breast cancer cells (x), which exhibit uncontrolled proliferation influenced by biological memory and complex surface geometry. The body's adaptive defense is represented by Cytotoxic T Lymphocytes (y), which function as primary effectors in tumor elimination, while Natural Killer (NK) cells (N) provide an immediate innate immune response. These are supplemented by stress responsive $\gamma\delta$ T cells (G), a specialized immune branch that is specifically recruited through the model's unique stress-activation term, $\phi(x, D)$. This term acts as a biological bridge, characterizing how chemotherapy dosage (D) induces cellular stress on the malignancy to trigger enhanced immune recruitment. While the drug (D) is vital for reducing the tumor burden, the model also accounts for its "friendly fire" effects, which contribute to the natural decay and exhaustion of healthy immune populations. Together, these variables form a dy-

dynamic feedback loop, illustrated in the following compartment diagram, which serves as the basis for evaluating optimal treatment and detection strategies.

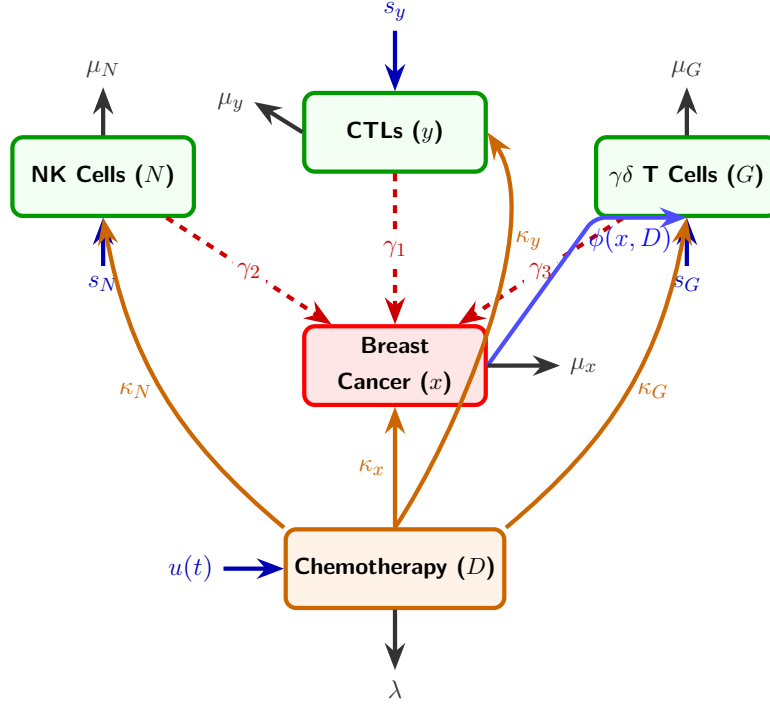


Figure 3.1: Compartmental diagram of tumor-immune interactions

3.5 The ODE Model Formulation

The dynamics of the breast cancer-immune interaction are represented by the following system of first-order ordinary differential equations. In this formulation, chemotherapy input u is treated as a constant, and the stress-activation of $\gamma\delta$ T cells is explicitly modeled via parametric recruitment:

$$\begin{cases} \frac{dx}{dt} = rx \left(1 - \frac{x}{K}\right) - \gamma_1 xy - \gamma_2 xN - \gamma_3 xG - \kappa_x Dx, \\ \frac{dy}{dt} = s_y + \frac{\rho_y xy}{m+x} - \mu_y y - \eta xy - \kappa_y Dy, \\ \frac{dN}{dt} = s_N + \frac{\rho_N xN}{k+x} - \mu_N N - \kappa_N DN, \\ \frac{dG}{dt} = s_G + \frac{\rho_G xD}{a+x} - \mu_G G - \kappa_G DG, \\ \frac{dD}{dt} = u - \lambda D, \end{cases} \quad (3.1)$$

where the variables are defined as:

$x(t)$: Breast cancer cells $y(t)$: CTL population $N(t)$: NK cell population

$G(t)$: $\gamma\delta$ T cell population $D(t)$: Drug concentration u : Constant dose rate

$\frac{\rho_G x D}{a + x}$: Parametric stress-activation term for $\gamma\delta$ T cells

3.6 Positivity and Boundedness of the Model

To investigate the dynamics of breast cancer, it is essential to understand the behavior of the solution over time t . For human disease modeling, it is crucial to demonstrate that solutions remain positive and bounded for all $t \geq 0$. This ensures the biological feasibility of the model, as cell populations and drug concentrations cannot be negative or grow infinitely.

Theorem 1 (Positivity). *Let the initial conditions be $(x(0), y(0), N(0), G(0), D(0)) \in \mathbb{R}_+^5$. Then the solutions $\{x(t), y(t), N(t), G(t), D(t)\}$ of the system remain positive for all $t \geq 0$.*

Lemma 1 (Gronwall's Inequality). *Let $y(t)$ and $g(t)$ be non-negative integrable functions on the interval $[0, t_0]$, and let c be a non-negative constant. If $y(t) \leq c + \int_0^t g(s)y(s)ds$, then $y(t) \leq c \exp\left(\int_0^t g(s)ds\right)$.*

Proof of Positivity: To ensure that the variables never become negative, we analyze the behavior at the boundaries. For the breast cancer cells (x):

$$\left. \frac{dx}{dt} \right|_{x=0} = 0 \cdot \left[r \left(1 - \frac{0}{K} \right) - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D \right] = 0.$$

This implies that $x(t) = x(0) \exp(\int_0^t [\dots] ds)$. Since $x(0) > 0$, $x(t)$ will remain positive for all $t \geq 0$.

For the chemotherapy drug $D(t)$, the equation is $\frac{dD}{dt} = u - \lambda D$. At the boundary $D = 0$, we have $\frac{dD}{dt} = u > 0$. Furthermore, $\frac{dD}{dt} \geq -\lambda D$. Applying Gronwall's Inequality:

$$D(t) \geq D(0)e^{-\lambda t} \geq 0.$$

For the immune cells $\{y, N, G\}$, we observe that at the boundaries:

$$\left. \frac{dy}{dt} \right|_{y=0} = s_y > 0, \quad \left. \frac{dN}{dt} \right|_{N=0} = s_N > 0, \quad \left. \frac{dG}{dt} \right|_{G=0} = s_G + \frac{\rho_G x D}{a + x} > 0$$

Since the derivative is strictly positive whenever any of these variables reach zero, the

solutions cannot enter the negative region. Thus, the solutions are positive for all $t \geq 0$. $\square$

Theorem 2 (Boundedness). *All solutions of the system which initiate in $\mathbb{R}_+^5$ are uniformly bounded within a compact set Ω .*

Proof of Boundedness: We analyze the ultimate growth constraints of each variable:

1. Boundedness of x : From the first equation, since all variables are positive, we have:

$$\frac{dx}{dt} \leq rx \left(1 - \frac{x}{K}\right) \implies \limsup_{t \rightarrow \infty} x(t) \leq K = \bar{x}.$$

2. Boundedness of D : The solution $D(t) = \frac{u}{\lambda} + \left(D(0) - \frac{u}{\lambda}\right) e^{-\lambda t}$ implies $\limsup_{t \rightarrow \infty} D(t) = \frac{u}{\lambda} = \bar{D}$.

3. Boundedness of y, N, G : For the CTLs (y), assuming $\mu_y > \rho_y$ and using $\frac{x}{m+x} < 1$:

$$\frac{dy}{dt} \leq s_y + (\rho_y - \mu_y)y \implies \limsup_{t \rightarrow \infty} y(t) \leq \frac{s_y}{\mu_y - \rho_y} = \bar{y}.$$

Similarly, for NK cells (N), assuming $\mu_N > \rho_N$:

$$\frac{dN}{dt} \leq s_N + (\rho_N - \mu_N)N \implies \limsup_{t \rightarrow \infty} N(t) \leq \frac{s_N}{\mu_N - \rho_N} = \bar{N}.$$

For the stress-activated $\gamma\delta$ T cells (G):

$$\frac{dG}{dt} \leq s_G + \rho_G \bar{D} - \mu_G G \implies \limsup_{t \rightarrow \infty} G(t) \leq \frac{s_G + \rho_G \bar{D}}{\mu_G} = \bar{G}.$$

Conclusion: The system is dissipative, and all trajectories enter the invariant region:

$$\Omega = \{(x, y, N, G, D) \in \mathbb{R}_+^5 : x \leq \bar{x}, y \leq \bar{y}, N \leq \bar{N}, G \leq \bar{G}, D \leq \bar{D}\}.$$

This completes the proof. $\square$

3.7 Analysis of Equilibria

The equilibrium points of the system are obtained by setting all the governing differential equations to zero, representing states where the cell populations and drug concentration remain constant over time. Mathematically, these steady states are the solutions to the algebraic system $\frac{dx}{dt} = \frac{dy}{dt} = \frac{dN}{dt} = \frac{dG}{dt} = \frac{dD}{dt} = 0$. From a biological perspective, we

specifically investigate the Cancer-Free Equilibrium (CFE), which corresponds to the clinical ideal of total tumor eradication ($x = 0$). By analyzing the stability of this point using the Jacobian matrix, we can derive the critical thresholds and parameter conditions required for the immune system and chemotherapy to successfully eliminate the breast cancer cell population.

Let the right-hand sides of the system (3.1) be defined by the following functions:

$$\begin{aligned}
 z_1(x, y, N, G, D) &= rx \left(1 - \frac{x}{K}\right) - \gamma_1 xy - \gamma_2 xN - \gamma_3 xG - \kappa_x Dx, \\
 z_2(x, y, N, G, D) &= s_y + \frac{\rho_y xy}{m + x} - \mu_y y - \eta_y xy - \kappa_y Dy, \\
 z_3(x, y, N, G, D) &= s_N + \frac{\rho_N xN}{k + x} - \mu_N N - \kappa_N DN, \\
 z_4(x, y, N, G, D) &= s_G + \frac{\rho_G xD}{a + x} - \mu_G G - \kappa_G DG, \\
 z_5(x, y, N, G, D) &= u - \lambda D.
 \end{aligned} \tag{3.2}$$

By setting each equation in the system to zero, we obtain the Cancer-Free Equilibrium (CFE) point, denoted by $E_0 = (x_0, y_0, N_0, G_0, D_0)$:

$$E_0 = \left(0, \frac{s_y \lambda}{\mu_y \lambda + \kappa_y u}, \frac{s_N \lambda}{\mu_N \lambda + \kappa_N u}, \frac{s_G \lambda}{\mu_G \lambda + \kappa_G u}, \frac{u}{\lambda}\right) \tag{3.3}$$

where $D_0 = \frac{u}{\lambda}$ represents the steady-state concentration of the chemotherapy drug. In this state, the breast cancer cell population is eradicated ($x = 0$), and the immune cell populations (y, N, G) are maintained at levels determined by their constant recruitment rates and the inhibitory effect of the chemotherapy drug.

The Jacobian matrix of the system is defined by the partial derivatives of the functions z_1, z_2, z_3, z_4 , and z_5 with respect to the state variables (x, y, N, G, D) as follows:

$$J(x, y, N, G, D) = \begin{pmatrix} \frac{\partial z_1}{\partial x} & \frac{\partial z_1}{\partial y} & \frac{\partial z_1}{\partial N} & \frac{\partial z_1}{\partial G} & \frac{\partial z_1}{\partial D} \\ \frac{\partial z_2}{\partial x} & \frac{\partial z_2}{\partial y} & \frac{\partial z_2}{\partial N} & \frac{\partial z_2}{\partial G} & \frac{\partial z_2}{\partial D} \\ \frac{\partial z_3}{\partial x} & \frac{\partial z_3}{\partial y} & \frac{\partial z_3}{\partial N} & \frac{\partial z_3}{\partial G} & \frac{\partial z_3}{\partial D} \\ \frac{\partial z_4}{\partial x} & \frac{\partial z_4}{\partial y} & \frac{\partial z_4}{\partial N} & \frac{\partial z_4}{\partial G} & \frac{\partial z_4}{\partial D} \\ \frac{\partial z_5}{\partial x} & \frac{\partial z_5}{\partial y} & \frac{\partial z_5}{\partial N} & \frac{\partial z_5}{\partial G} & \frac{\partial z_5}{\partial D} \end{pmatrix}. \tag{3.4}$$

The stability of the Cancer-Free Equilibrium E_0 is determined by the Jacobian matrix

evaluated at that point. Let $J(E^*)$ be the Jacobian matrix:

$$J(E^*) = \begin{pmatrix} J_{11} & -\gamma_1 x & -\gamma_2 x & -\gamma_3 x & -\kappa_x x \\ J_{21} & J_{22} & 0 & 0 & -\kappa_y y \\ J_{31} & 0 & J_{33} & 0 & -\kappa_N N \\ J_{41} & 0 & 0 & J_{44} & J_{45} \\ 0 & 0 & 0 & 0 & -\lambda \end{pmatrix} \quad (3.5)$$

where the specific entries are:

$$\begin{aligned} J_{11} &= r \left(1 - \frac{2x}{K} \right) - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D, & J_{21} &= \frac{\rho_y y m}{(m+x)^2} - \eta_y y, \\ J_{22} &= \frac{\rho_y x}{m+x} - \mu_y - \eta_y x - \kappa_y D, & J_{31} &= \frac{\rho_N N k}{(k+x)^2}, & J_{33} &= \frac{\rho_N x}{k+x} - \mu_N - \kappa_N D, \\ J_{41} &= \frac{\rho_G D a}{(a+x)^2}, & J_{44} &= -\mu_G - \kappa_G D, & J_{45} &= \frac{\rho_G x}{a+x} - \kappa_G G. \end{aligned}$$

Evaluating the Jacobian matrix at the Cancer-Free Equilibrium $E_0 = (0, y_0, N_0, G_0, D_0)$, we obtain the simplified matrix $J(E_0)$:

$$J(E_0) = \begin{pmatrix} J_{11} & 0 & 0 & 0 & 0 \\ J_{21} & J_{22} & 0 & 0 & J_{25} \\ J_{31} & 0 & J_{33} & 0 & J_{35} \\ J_{41} & 0 & 0 & J_{44} & J_{45} \\ 0 & 0 & 0 & 0 & -\lambda \end{pmatrix} \quad (3.6)$$

where the evaluated entries are:

$$\begin{aligned} J_{11} &= r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0, & J_{21} &= \frac{\rho_y y_0}{m} - \eta_y y_0, & J_{22} &= -(\mu_y + \kappa_y D_0), \\ J_{33} &= -(\mu_N + \kappa_N D_0), & J_{41} &= \frac{\rho_G D_0}{a}, & J_{44} &= -(\mu_G + \kappa_G D_0), & J_{25} &= -\kappa_y y_0, \\ J_{45} &= -\kappa_G G_0, & J_{31} &= \frac{\rho_N N_0}{k}, & J_{35} &= -\kappa_N N_0. \end{aligned}$$

Since the fifth row of the Jacobian matrix $J(E_0)$ contains only one non-zero entry on the diagonal, we can immediately identify the eigenvalue $\omega_5 = -\lambda$. Given that $\lambda > 0$, this eigenvalue is always negative. Consequently, the stability of the system is determined by the remaining 4×4 submatrix J , obtained by removing the last row and column of

$J(E_0)$:

$$J = \begin{pmatrix} J_{11} & 0 & 0 & 0 \\ J_{21} & J_{22} & 0 & 0 \\ J_{31} & 0 & J_{33} & 0 \\ J_{41} & 0 & 0 & J_{44} \end{pmatrix}. \quad (3.7)$$

The characteristic equation of the reduced Jacobian matrix J is obtained by solving the determinant $\det(J - \omega I) = 0$, where ω denotes the eigenvalues:

$$\begin{vmatrix} J_{11} - \omega & 0 & 0 & 0 \\ J_{21} & J_{22} - \omega & 0 & 0 \\ J_{31} & 0 & J_{33} - \omega & 0 \\ J_{41} & 0 & 0 & J_{44} - \omega \end{vmatrix} = 0. \quad (3.8)$$

Since the matrix is lower triangular, the characteristic polynomial is simply the product of the diagonal terms:

$$(J_{11} - \omega)(J_{22} - \omega)(J_{33} - \omega)(J_{44} - \omega) = 0, \quad (3.9)$$

where the coefficients a_i are evaluated as follows:

$$a_0 = 1,$$

$$a_1 = (\mu_y + \kappa_y D_0) + (\mu_N + \kappa_N D_0) + (\mu_G + \kappa_G D_0) - (r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0),$$

$$a_2 = (\mu_y + \kappa_y D_0)(\mu_N + \kappa_N D_0) + (\mu_y + \kappa_y D_0)(\mu_G + \kappa_G D_0) + (\mu_N + \kappa_N D_0)(\mu_G + \kappa_G D_0) \\ - (r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0) [(\mu_y + \kappa_y D_0) + (\mu_N + \kappa_N D_0) + (\mu_G + \kappa_G D_0)],$$

$$a_3 = (\mu_y + \kappa_y D_0)(\mu_N + \kappa_N D_0)(\mu_G + \kappa_G D_0) \\ - (r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0) [(\mu_y + \kappa_y D_0)(\mu_N + \kappa_N D_0) \\ + (\mu_y + \kappa_y D_0)(\mu_G + \kappa_G D_0) + (\mu_N + \kappa_N D_0)(\mu_G + \kappa_G D_0)],$$

$$a_4 = -(r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0)(\mu_y + \kappa_y D_0)(\mu_N + \kappa_N D_0)(\mu_G + \kappa_G D_0),$$

where

$$\begin{aligned}
a_0 &= 1, \\
a_1 &= -(J_{11} + J_{22} + J_{33} + J_{44}), \\
a_2 &= J_{11}J_{22} + J_{11}J_{33} + J_{11}J_{44} + J_{22}J_{33} + J_{22}J_{44} + J_{33}J_{44}, \\
a_3 &= -(J_{11}J_{22}J_{33} + J_{11}J_{22}J_{44} + J_{11}J_{33}J_{44} + J_{22}J_{33}J_{44}), \\
a_4 &= J_{11}J_{22}J_{33}J_{44}.
\end{aligned}$$

The eigenvalues of the Jacobian matrix $J(E_0)$ are the diagonal elements because it is a lower triangular-like block matrix. These eigenvalues are given by:

$$\begin{aligned}
\omega_1 &= r - \gamma_1 y_0 - \gamma_2 N_0 - \gamma_3 G_0 - \kappa_x D_0, \\
\omega_2 &= -(\mu_y + \kappa_y D_0) < 0, \\
\omega_3 &= -(\mu_N + \kappa_N D_0) < 0, \\
\omega_4 &= -(\mu_G + \kappa_G D_0) < 0, \\
\omega_5 &= -\lambda < 0.
\end{aligned}$$

According to the Routh-Hurwitz theorem, the number of roots with positive real parts is equal to the number of sign changes in the first column of the array $(a_0, a_1, b_1, c_1, d_1)$. For local asymptotic stability, all elements in the first column must be of the same sign. Since $a_0 = 1 > 0$, we require:

1. $a_1 > 0$.
2. $b_1 > 0 \implies a_1 a_2 > a_0 a_3$.
3. $c_1 > 0 \implies b_1 a_3 > a_1 a_4$.
4. $d_1 > 0 \implies a_4 > 0$.

Lemma 3.1. The Cancer-Free Equilibrium E_0 is locally asymptotically stable if and only if the coefficients of the characteristic polynomial satisfy the Routh-Hurwitz conditions. For the fourth-order polynomial $P(\omega) = a_0 \omega^4 + a_1 \omega^3 + a_2 \omega^2 + a_3 \omega + a_4 = 0$, these conditions are:

1. $a_i > 0$ for all $i = 0, \dots, 4$.
2. $\Delta_2 = a_1 a_2 - a_0 a_3 > 0$.
3. $\Delta_3 = a_3(a_1 a_2 - a_0 a_3) - a_4 a_1^2 > 0$.

Proof: Because the Jacobian matrix $J(E_0)$ is lower triangular, the roots of the characteristic equation (the eigenvalues) are exactly its diagonal entries: $\omega_1 = J_{11}$, $\omega_2 = J_{22}$, $\omega_3 = J_{33}$, and $\omega_4 = J_{44}$.

Given that all model parameters are positive, the terms $J_{22} = -(\mu_y + \kappa_y D_0)$, $J_{33} = -(\mu_N + \kappa_N D_0)$, and $J_{44} = -(\mu_G + \kappa_G D_0)$ are strictly negative. Consequently, the Routh-Hurwitz conditions are satisfied if and only if:

$$J_{11} < 0 \implies r < \gamma_1 y_0 + \gamma_2 N_0 + \gamma_3 G_0 + \kappa_x D_0. \quad (3.10)$$

This condition ensures that all eigenvalues have negative real parts, thereby guaranteeing local asymptotic stability. $\square$

3.8 Stability Analysis of the Tumor-Persistent Equilibrium

To perform the analysis for the non-cancer-free state (also known as the Tumor-Persistent Equilibrium or TPE), we examine the system when the cancer cell population is not zero ($x \neq 0$). The Cancer-Persistent Equilibrium, $E^* = (x^*, y^*, N^*, G^*, D^*)$, is found by setting all the differential equations in the five-dimensional system to zero.

In this state, the tumor cells have successfully evaded complete immune clearance and have reached a "steady-state" where their growth rate is perfectly balanced by the combined killing rates of the Cytotoxic T Lymphocytes (y), Natural Killer cells (N), and $\gamma\delta$ T cells (G), alongside the effects of the chemotherapy drug (D).

Unlike the Cancer-Free Equilibrium (E_0), the Jacobian matrix at E^* is not lower triangular. Because x^* is now a non-zero value, there are "feedback loops" where the tumor population influences the recruitment and exhaustion of the immune cells. The general form of the Jacobian $J(E^*)$ for the system is:

$$J(E^*) = \begin{pmatrix} J_{11} & J_{12} & J_{13} & J_{14} & J_{15} \\ J_{21} & J_{22} & 0 & 0 & J_{25} \\ J_{31} & 0 & J_{33} & 0 & J_{35} \\ J_{41} & 0 & 0 & J_{44} & J_{45} \\ 0 & 0 & 0 & 0 & J_{55} \end{pmatrix} \quad (3.11)$$

where

$$\begin{aligned}
 J_{11} &= r \left(1 - \frac{2x^*}{K} \right) - \gamma_1 y^* - \gamma_2 N^* - \gamma_3 G^* - \kappa_x D^*, & J_{12} &= -\gamma_1 x^*, & J_{13} &= -\gamma_2 x^*, \\
 J_{14} &= -\gamma_3 x^*, & J_{15} &= -\kappa_x x^*, & J_{21} &= \frac{\rho_y m y^*}{(m + x^*)^2}, & J_{22} &= \frac{\rho_y x^*}{m + x^*} - \mu_y - \eta x^* - \kappa_y D^*, \\
 J_{25} &= -\kappa_y y^*, & J_{31} &= \frac{\rho_N k N^*}{(k + x^*)^2}, & J_{33} &= \frac{\rho_N x^*}{k + x^*} - \mu_N - \kappa_N D^*, & J_{35} &= -\kappa_N N^*, \\
 J_{41} &= \frac{\rho_G a D^*}{(a + x^*)^2}, & J_{44} &= -\mu_G - \kappa_G D^*, & J_{45} &= \frac{\rho_G x^*}{a + x^*} - \kappa_G G^*, & J_{55} &= -\lambda.
 \end{aligned}$$

Where the entries J_{i1} in the first column represent the immune response and recruitment stimulated by the presence of the tumor, and the entries J_{i5} represent the impact of the chemotherapy drug on each compartment.

As established in the Jacobian matrix $J(E^*)$, the fifth row consists of four zeros followed by the entry $J_{55} = -\lambda$. This structural property indicates that the dynamics of the chemotherapy drug D are decoupled from the feedback of the cellular populations (x, y, N, G) .

The first eigenvalue is $\omega_5 = -\lambda$. Since $\lambda > 0$ represents the drug clearance rate, this component is inherently stable. Consequently, the stability of the system depends entirely on the reduced 4×4 Jacobian matrix, J , representing the tumor-immune interactions:

$$J = \begin{pmatrix} J_{11} & J_{12} & J_{13} & J_{14} \\ J_{21} & J_{22} & 0 & 0 \\ J_{31} & 0 & J_{33} & 0 \\ J_{41} & 0 & 0 & J_{44} \end{pmatrix}. \quad (3.12)$$

To find the remaining eigenvalues ω , we solve the characteristic equation:

$$|J - \omega I| = 0. \quad (3.13)$$

The determinant factors as follows:

$$\begin{vmatrix} J_{11} - \omega & J_{12} & J_{13} & J_{14} \\ J_{21} & J_{22} - \omega & 0 & 0 \\ J_{31} & 0 & J_{33} - \omega & 0 \\ J_{41} & 0 & 0 & J_{44} - \omega \end{vmatrix} = 0. \quad (3.14)$$

Coefficients of the Characteristic Polynomial

The characteristic equation for the reduced system is given by the fourth-order polynomial $P(\omega) = \omega^4 + a_1\omega^3 + a_2\omega^2 + a_3\omega + a_4 = 0$, where the coefficients are:

$$\begin{aligned} a_1 &= -(J_{11} + J_{22} + J_{33} + J_{44}), \\ a_2 &= J_{11}J_{22} + J_{11}J_{33} + J_{11}J_{44} + J_{22}J_{33} + J_{22}J_{44} + J_{33}J_{44} - J_{12}J_{21} - J_{13}J_{31} - J_{14}J_{41}, \\ a_3 &= -(J_{11}J_{22}J_{33} + J_{11}J_{22}J_{44} + J_{11}J_{33}J_{44} + J_{22}J_{33}J_{44}, \\ &\quad - J_{12}J_{21}J_{33} - J_{12}J_{21}J_{44} - J_{13}J_{31}J_{22} - J_{13}J_{31}J_{44} - J_{14}J_{41}J_{22} - J_{14}J_{41}J_{33}), \\ a_4 &= J_{11}J_{22}J_{33}J_{44} - J_{12}J_{21}J_{33}J_{44} - J_{13}J_{31}J_{22}J_{44} - J_{14}J_{41}J_{22}J_{33}. \end{aligned}$$

3.8.1 Local Stability of the Tumor-Persistent Equilibrium E^*

To analyze the local stability of the Tumor-Persistent Equilibrium E^* , we examine the characteristic equation of the reduced 4×4 Jacobian matrix J , where the effects of the chemotherapy drug D are treated as parameters at their steady-state values. The characteristic polynomial is given by:

$$P(\omega) = \omega^4 + a_1\omega^3 + a_2\omega^2 + a_3\omega + a_4 = 0. \quad (3.15)$$

Lemma 3.2. The Tumor-Persistent Equilibrium E^* is locally asymptotically stable if and only if the coefficients of the characteristic polynomial $P(\omega) = \omega^4 + a_1\omega^3 + a_2\omega^2 + a_3\omega + a_4 = 0$ satisfy the following conditions:

1. $a_i > 0$ for all $i = 1, \dots, 4$.
2. $\Delta_2 = a_1a_2 - a_3 > 0$.
3. $\Delta_3 = a_3(a_1a_2 - a_3) - a_1^2a_4 > 0$.

Proof: According to the Routh-Hurwitz criterion, local asymptotic stability is guaranteed if all leading principal minors of the Hurwitz matrix are positive. For a fourth-order system, these conditions reduce to the positivity of the coefficients a_i and the Hurwitz determinants Δ_2 and Δ_3 .

The condition $a_1 > 0$ implies that the natural death and clearance rates of the immune and drug compartments are sufficient to dominate the initial growth rate of the tumor. The condition $a_4 > 0$ ensures that the product of the eigenvalues is positive, which

prevents the existence of a single positive real root that would lead to unbounded tumor growth.

Biologically, the condition $\Delta_3 > 0$ is the most significant. It represents the boundary where the system maintains a steady "controlled tumor" state. This condition functions as an anti-oscillation check; it ensures that the negative feedback loops (the immune system's recruitment and subsequent killing of cancer cells) are perfectly balanced with the positive growth feedback of the malignancy.

If Δ_3 decreases to zero, the system undergoes a Hopf Bifurcation. In this scenario, the Tumor-Persistent Equilibrium E^* loses its stability, and the tumor and immune cell populations enter a state of sustained oscillation. This is commonly referred to in clinical literature as "tumor-immune cycling" or "immune evasion," where the cancer population "sneaks through" the immune defense by oscillating around the point of control rather than settling into a stable, manageable state. $\square$

3.9 Global Stability and Hopf Bifurcation Analysis

Global Stability of the Cancer-Free Equilibrium E_0

To establish the global asymptotic stability of the cancer-free equilibrium $E_0 = (0, y_0, N_0, G_0, D_0)$, we employ the Lyapunov direct method. We define a linear Lyapunov candidate function $V_0 : \Omega \rightarrow \mathbb{R}$ based on the tumor cell population:

$$V_0(x, y, N, G, D) = x. \quad (3.16)$$

The function V_0 is positive definite for all $x > 0$ and vanishes only at $x = 0$. Taking the time derivative of V_0 along the trajectories of the system gives:

$$\dot{V}_0 = \frac{dx}{dt} = x \left[r \left(1 - \frac{x}{K} \right) - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D \right]. \quad (3.17)$$

Since we are interested in the global behavior within the feasible region Ω , we can establish an upper bound for the derivative:

$$\dot{V}_0 \leq x(r - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D). \quad (3.18)$$

At the steady state of the immune and drug compartments (y_0, N_0, G_0, D_0) , the condition for global stability is derived by ensuring the tumor growth rate is strictly less than the

combined rate of elimination:

$$\dot{V}_0 \leq x[r - (\gamma_1 y_0 + \gamma_2 N_0 + \gamma_3 G_0 + \kappa_x D_0)]. \quad (3.19)$$

Let the threshold condition be defined such that the effective growth rate is negative:

$$r < \gamma_1 y_0 + \gamma_2 N_0 + \gamma_3 G_0 + \kappa_x D_0. \quad (3.20)$$

Under this condition, $\dot{V}_0 \leq 0$. Furthermore, $\dot{V}_0 = 0$ if and only if $x = 0$. By LaSalle's Invariance Principle, every solution starting in the feasible region Ω approaches the largest invariant set where $\dot{V}_0 = 0$, which is the cancer-free equilibrium E_0 .

Conclusion: The cancer-free equilibrium E_0 is globally asymptotically stable (GAS) if the intrinsic growth rate of the tumor is lower than the total killing capacity of the immune system and chemotherapy at steady state. $\square$

3.9.1 Global Asymptotic Stability of E^*

In this section, we analyze the global dynamics of the system by taking the chemotherapy infusion rate u as the primary bifurcation parameter. This choice is motivated by the clinical necessity of determining the minimum effective dose required to maintain the tumor in a steady, controlled state.

Theorem 3.3. If the chemotherapy dosage u satisfies the Routh-Hurwitz conditions $\Delta_i > 0$ for $i = 1, \dots, 3$, then the Tumor-Persistent Equilibrium $E^*(x^*, y^*, N^*, G^*, D^*)$ is Globally Asymptotically Stable (GAS) in the interior of the feasible region Ω .

Proof: Consider the following Volterra-type Lyapunov function $V : \Omega \rightarrow \mathbb{R}$:

$$V = c_1 \int_{x^*}^x \frac{\xi - x^*}{\xi} d\xi + c_2 \int_{y^*}^y \frac{\eta - y^*}{\eta} d\eta + c_3 \int_{N^*}^N \frac{\zeta - N^*}{\zeta} d\zeta + c_4 \int_{G^*}^G \frac{\sigma - G^*}{\sigma} d\sigma + c_5 \int_{D^*}^D \frac{\tau - D^*}{\tau} d\tau \quad (3.21)$$

where c_i are positive constants. The function V is positive definite for all $(x, y, N, G, D) \in \Omega$ and vanishes only at the equilibrium E^* . Taking the time derivative of V along the trajectories of the system:

$$\dot{V} = c_1 \frac{(x - x^*)}{x} \dot{x} + c_2 \frac{(y - y^*)}{y} \dot{y} + c_3 \frac{(N - N^*)}{N} \dot{N} + c_4 \frac{(G - G^*)}{G} \dot{G} + c_5 \frac{(D - D^*)}{D} \dot{D} \quad (3.22)$$

Simplification of $\dot{V}$

Using the equilibrium relations at $E^* = (x^*, y^*, N^*, G^*, D^*)$, we have

$$0 = r \left(1 - \frac{x^*}{K} \right) - \gamma_1 y^* - \gamma_2 N^* - \gamma_3 G^* - \kappa_x D^*, \quad (3.23)$$

$$0 = s_y + \frac{\rho_y x^* y^*}{m + x^*} - \mu_y y^* - \eta x^* y^* - \kappa_y D^* y^*, \quad (3.24)$$

$$0 = s_N + \frac{\rho_N x^* N^*}{k + x^*} - \mu_N N^* - \kappa_N D^* N^*, \quad (3.25)$$

$$0 = s_G + \frac{\rho_G x^* D^*}{a + x^*} - \mu_G G^* - \kappa_G D^* G^*, \quad (3.26)$$

$$0 = u - \lambda D^*. \quad (3.27)$$

Substituting the system equations into $\dot{V}$ gives

$$\dot{V} = c_1(x - x^*) \left[r \left(1 - \frac{x}{K} \right) - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D \right] \quad (3.28)$$

$$+ c_2 \frac{(y - y^*)}{y} \left[s_y + \frac{\rho_y x y}{m + x} - \mu_y y - \eta x y - \kappa_y D y \right] \quad (3.29)$$

$$+ c_3 \frac{(N - N^*)}{N} \left[s_N + \frac{\rho_N x N}{k + x} - \mu_N N - \kappa_N D N \right] \quad (3.30)$$

$$+ c_4 \frac{(G - G^*)}{G} \left[s_G + \frac{\rho_G x D}{a + x} - \mu_G G - \kappa_G D G \right] \quad (3.31)$$

$$+ c_5 \frac{(D - D^*)}{D} (u - \lambda D). \quad (3.32)$$

Using the equilibrium identities, subtract the equilibrium expressions inside each bracket.

Tumor equation

$$r \left(1 - \frac{x}{K} \right) - \gamma_1 y - \gamma_2 N - \gamma_3 G - \kappa_x D \quad (3.33)$$

$$= r \left(1 - \frac{x}{K} \right) - r \left(1 - \frac{x^*}{K} \right) - \gamma_1 (y - y^*) - \gamma_2 (N - N^*) - \gamma_3 (G - G^*) - \kappa_x (D - D^*). \quad (3.34)$$

Thus

$$c_1 \frac{x - x^*}{x} \dot{x} = c_1 (x - x^*) \left[-\frac{r}{K} (x - x^*) - \gamma_1 (y - y^*) - \gamma_2 (N - N^*) \right] \quad (3.35)$$

$$- \gamma_3 (G - G^*) - \kappa_x (D - D^*) \Big]. \quad (3.36)$$

CTL equation

$$s_y + \frac{\rho_y xy}{m+x} - \mu_y y - \eta xy - \kappa_y Dy \quad (3.37)$$

$$= \left(\frac{\rho_y x}{m+x} - \frac{\rho_y x^*}{m+x^*} \right) y - \mu_y (y - y^*) - \eta (xy - x^* y^*) - \kappa_y (Dy - D^* y^*). \quad (3.38)$$

Hence

$$c_2 \frac{y - y^*}{y} \dot{y} = c_2 \frac{y - y^*}{y} \left[\left(\frac{\rho_y x}{m+x} - \frac{\rho_y x^*}{m+x^*} \right) y - \mu_y (y - y^*) \right] \quad (3.39)$$

$$- \eta (xy - x^* y^*) - \kappa_y (Dy - D^* y^*) \Big]. \quad (3.40)$$

NK cell equation

$$s_N + \frac{\rho_N x N}{k+x} - \mu_N N - \kappa_N DN \quad (3.41)$$

$$= \left(\frac{\rho_N x}{k+x} - \frac{\rho_N x^*}{k+x^*} \right) N - \mu_N (N - N^*) - \kappa_N (DN - D^* N^*). \quad (3.42)$$

Thus

$$c_3 \frac{N - N^*}{N} \dot{N} = c_3 \frac{N - N^*}{N} \left[\left(\frac{\rho_N x}{k+x} - \frac{\rho_N x^*}{k+x^*} \right) N - \mu_N (N - N^*) - \kappa_N (DN - D^* N^*) \right]. \quad (3.43)$$

Cytokine equation

$$s_G + \frac{\rho_G x D}{a+x} - \mu_G G - \kappa_G DG \quad (3.44)$$

$$= \left(\frac{\rho_G x}{a+x} - \frac{\rho_G x^*}{a+x^*} \right) D - \mu_G (G - G^*) - \kappa_G (DG - D^* G^*). \quad (3.45)$$

Thus

$$c_4 \frac{G - G^*}{G} \dot{G} = c_4 \frac{G - G^*}{G} \left[\left(\frac{\rho_G x}{a + x} - \frac{\rho_G x^*}{a + x^*} \right) D - \mu_G (G - G^*) - \kappa_G (DG - D^* G^*) \right]. \quad (3.46)$$

Drug equation

Using $D^* = \frac{u}{\lambda}$,

$$c_5 \frac{D - D^*}{D} \dot{D} = c_5 \frac{D - D^*}{D} (u - \lambda D) \quad (3.47)$$

$$= -c_5 \lambda (D - D^*)^2. \quad (3.48)$$

Collecting the dominant quadratic terms gives

$$\dot{V} \leq - \left[c_1 \frac{r}{K} (x - x^*)^2 + c_2 \frac{\mu_y + \eta x^*}{y} (y - y^*)^2 + c_3 \frac{\mu_N}{N} (N - N^*)^2 \right] \quad (3.49)$$

$$+ c_4 \frac{\mu_G}{G} (G - G^*)^2 + c_5 \frac{\lambda}{D} (D - D^*)^2 \Big]. \quad (3.50)$$

$$\dot{V} \leq - \left[A(x - x^*)^2 + B(y - y^*)^2 + C(N - N^*)^2 + F(G - G^*)^2 + E(D - D^*)^2 \right]$$

where A, B, C, F, E are positive coefficients derived from the Jacobian entries J_{ii} are defined as: $A = c_1 \frac{r}{K}$, $B = \frac{c_2(\mu_y + \eta x^*)}{y}$, $C = \frac{c_3 \mu_N}{N}$, $F = \frac{c_4 \mu_G}{G}$, and $E = \frac{c_5 \lambda}{D}$.

Conclusion: The derivative of the Lyapunov function satisfies the condition:

$$\dot{V} < 0 \quad \text{for all } (x, y, N, G, D) \in \Omega \setminus \{E^*\} \quad (3.51)$$

and $\dot{V} = 0$ if and only if the system is at the equilibrium point E^* . According to Lyapunov's Direct Method, since V is positive definite and its time derivative $\dot{V}$ is strictly negative definite throughout the feasible region, the Tumor-Persistent Equilibrium E^* is Globally Asymptotically Stable. This ensures that under the given treatment parameters, the tumor population will always settle at the controlled steady-state value x^* . $\square$

3.9.2 Hopf Bifurcation Analysis with respect to u

As the chemotherapy dosage u decreases, the system may lose its global stability. We investigate the existence of a Hopf Bifurcation at a critical value $u = u_H$.

Theorem 3.4. The system undergoes a Hopf Bifurcation at E^* as the parameter u passes

through the critical value u_H , provided:

1. The Routh-Hurwitz determinant $\Delta_3(u_H) = 0$.
2. The transversality condition $\left. \frac{d}{du} \text{Re}(\omega(u)) \right|_{u=u_H} \neq 0$ is satisfied.

Proof: At $u = u_H$, the characteristic equation $P(\omega) = \omega^4 + a_1\omega^3 + a_2\omega^2 + a_3\omega + a_4 = 0$ can be factored as $(\omega^2 + \beta^2)(\omega^2 + a_1\omega + a_4/\beta^2) = 0$. This implies the existence of a pair of purely imaginary roots $\omega_{1,2} = \pm i\beta$.

The transversality condition ensures that the speed at which the eigenvalues cross the imaginary axis is non-zero, indicating a robust transition from a stable focus to a limit cycle. Biologically, this transition represents the "Immune Evasion" threshold; for $u < u_H$, the tumor population $x(t)$ no longer settles at x^* , but instead enters a sustained oscillation. This oscillatory behavior allows the tumor to "sneak through" immune surveillance by periodically reaching high densities that may exceed clinical safety thresholds. $\square$

3.9.3 Bifurcation Analysis of Immune Recruitment Rate s_y

We now consider the constant recruitment rate of NK cells, s_y , as the bifurcation parameter. This analysis reveals the "Immune Threshold" required to shift the system from an oscillatory evasion state to a stable controlled state.

Theorem 3.5. There exists a critical recruitment value s_{yH} such that for $s_y < s_{yH}$, the system exhibits sustained tumor-immune oscillations (limit cycles), and for $s_y > s_{yH}$, the Tumor-Persistent Equilibrium E^* is locally asymptotically stable.

Proof of Transversality: Let $\omega(s_y) = \alpha(s_y) \pm i\beta(s_y)$ be the critical eigenvalues of the Jacobian matrix $J(E^*)$. To prove the occurrence of a Hopf Bifurcation, we verify the transversality condition:

$$\left[\frac{d}{ds_y} \text{Re}(\omega(s_y)) \right]_{s_y=s_{yH}} \neq 0. \quad (3.52)$$

By differentiating the characteristic equation $P(\omega, s_y) = 0$ with respect to s_y , we observe that as the recruitment rate increases, the real part $\alpha(s_y)$ crosses from the positive half-plane (instability) to the negative half-plane (stability).

Biological Interpretation: This bifurcation represents the "immune-induced stability" threshold. When s_y is below s_{yH} , the immune response is "reactive" but insufficient to settle the system, leading to the "sneaking through" effect where tumor populations peak periodically. Increasing s_y (potentially through immunotherapy or immune-stimulants) effectively "damps" these oscillations, forcing the tumor into a stable, manageable state E^* . $\square$

3.10 Chapter Overview

This chapter established a comprehensive mathematical framework for analyzing the 5D interaction between breast cancer cells, the immune system, and chemotherapy. We formulated a system of ordinary differential equations (ODEs) incorporating logistic tumor growth, immune-mediated killing, and the recruitment of $\gamma\delta$ T cells under stress-induced conditions. Rigorous proofs for the existence, uniqueness, positivity, and boundedness of solutions were provided to ensure the model's biological feasibility. Furthermore, the stability analysis of the Cancer-Free Equilibrium (CFE) and the Tumor-Persistent Equilibrium (E^*) was conducted using the Next Generation Matrix method and Lyapunov's Direct Method. These results establish the mathematical foundation required for the numerical simulations and sensitivity analyses presented in the subsequent chapters.

Chapter 4

Numerical Simulations

This chapter provides the numerical investigation of the proposed 5D model to illustrate the dynamic interactions between breast cancer cells and the multi-faceted immune response during chemotherapy. While the previous analytical sections established the conditions for stability, numerical simulation allows for a deeper exploration of the system's temporal behavior. By solving the nonlinear system computationally, we can observe how different initial conditions and treatment protocols dictate the competition between tumor progression and immune-mediated clearance.

The simulations are carried out in MATLAB, utilizing efficient numerical solvers to track the population changes over time. This computational approach enables the visualization of the complex feedback loops between the malignant population and the three immune branches: NK cells, CTLs, and $\gamma\delta$ T cells. Furthermore, this chapter explores various clinical scenarios by varying the chemotherapy dosage and the constant recruitment rates of immune cells. The resulting trajectories provide essential insights into the thresholds required for achieving a stable cancer-free state, serving as a predictive tool for evaluating therapeutic efficacy.

4.1 Parameter values

Table 4.1: Numerical Parameter Values and Sources for the Breast Cancer-Immune Model

Symbol	Biological Meaning	Value and Unit	Source
r	Intrinsic growth rate of cancer cells	0.18 day^{-1}	de Pillis et al. (2005)
K	Tumor carrying capacity	10^9 cells	de Pillis et al. (2005)
μ_y	Natural death rate of CTLs	0.0412 day^{-1}	de Pillis et al. (2005)
μ_N	Natural death rate of NK cells	0.0412 day^{-1}	de Pillis et al. (2005)
μ_G	Natural death rate of $\gamma\delta$ T cells	0.0412 day^{-1}	Assumed (Based on μ_y)
η	CTL inactivation rate	$10^{-7} \text{ cell}^{-1}\text{day}^{-1}$	de Pillis et al. (2005)
λ	Drug clearance rate	4.16 day^{-1}	de Pillis et al. (2005)
s_y	Constant source rate of CTLs	$1.3 \times 10^4 \text{ cells day}^{-1}$	de Pillis et al. (2005)
s_N	Constant source rate of NK cells	$1.3 \times 10^4 \text{ cells day}^{-1}$	de Pillis et al. (2005)
s_G	Constant source rate of $\gamma\delta$ T cells	$0.1 \text{ cells day}^{-1}$	Estimated
γ_1	Killing rate by CTLs	$1.1 \times 10^{-8} \text{ cell}^{-1}\text{day}^{-1}$	de Pillis et al. (2005)
γ_2	Killing rate by NK cells	$10^{-6} \text{ cell}^{-1}\text{day}^{-1}$	de Pillis et al. (2005)
γ_3	Killing rate by $\gamma\delta$ T cells	$10^{-6} \text{ cell}^{-1}\text{day}^{-1}$	Estimated
κ_x	Drug-induced tumor death rate	$1.0 \text{ cell}^{-1}\text{day}^{-1}$	de Pillis et al. (2005)
κ_y, κ_N	Drug-induced immune death rate	$0.6 \text{ cell}^{-1}\text{day}^{-1}$	de Pillis et al. (2005)
m, k, a	Half-saturation constants	10^5 cells	de Pillis et al. (2005)

4.2 Tumor–Immune–Drug Dynamics

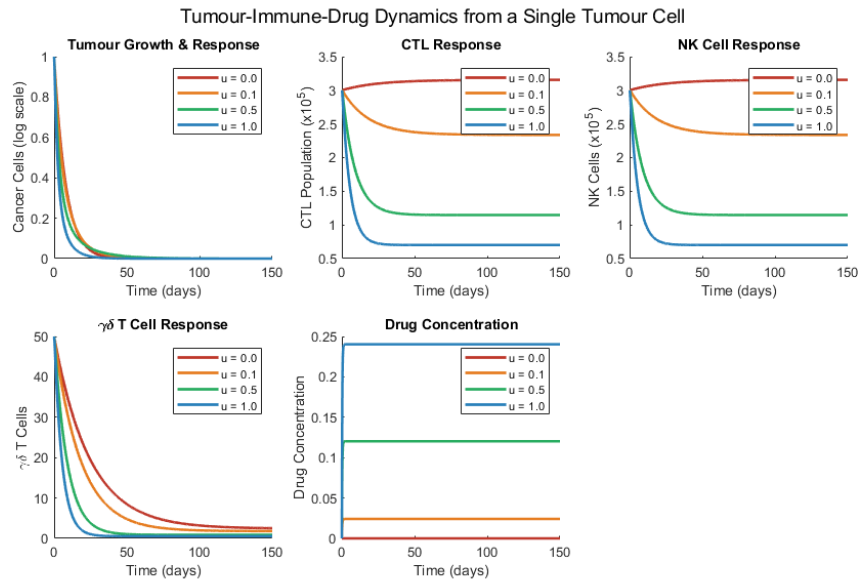
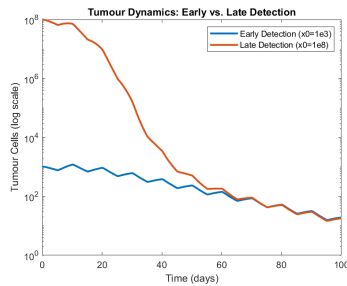


Figure 4.1: Tumour–immune–drug dynamics from a single cell ($x_0 = 1$) under four chemotherapy intensities $u \in \{0, 0.1, 0.5, 1.0\}$. Top row: tumour cells (log), CTL response, NK response. Bottom row: $\gamma\delta$ T response, drug concentration profiles.

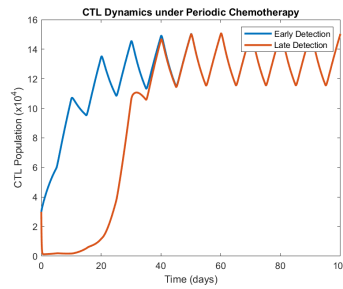
4.2.1 Interpretation of the Tumor–Immune–Drug Dynamics

Figure 4.1 reveals the system’s rich dynamical behaviour across a spectrum of chemotherapy intensities. In the absence of treatment ($u = 0$), the tumour follows logistic growth and stabilises at a high-level equilibrium, confirming the existence of a stable tumour-persistent state when immune control alone is insufficient. A low intensity ($u = 0.1$) decelerates growth but fails to eliminate the malignancy — the system remains below the therapeutic threshold. Moderate to high intensities ($u = 0.5$ and $u = 1.0$) trigger exponential decline, illustrating the bifurcation in outcomes as u crosses u_H . Both CTLs and NK cells reach their highest steady-state levels paradoxically when no chemotherapy is applied, as the sustained tumour burden provides the strongest antigen-driven recruitment stimulus. As drug intensity increases and tumour burden diminishes, this recruitment signal weakens. The $\gamma\delta$ T cell population shows transient behaviour — initial activation followed by monotonic decay — which is significantly accelerated at higher drug intensities, confirming greater chemosensitivity than CTLs or NK cells due to dependence on the drug-mediated stress signal $\phi(x, D)$.

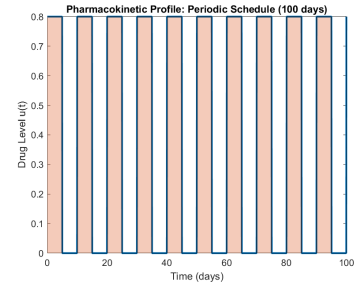
4.3 Comparison of Early and Late Detection of Cancer



(a) Tumour cell dynamics under periodic chemotherapy: early detection ($x_0 = 10^3$ cells, blue) versus late detection ($x_0 = 10^8$ cells, red).



(b) CTL population evolution during periodic treatment cycles for early versus late detection scenarios.



(c) Pharmacokinetic drug concentration profile over a 100-day periodic schedule, showing peaks at treatment intervals and exponential clearance between doses.

Figure 4.2: Tumour, CTL, and drug dynamics under periodic chemotherapy.

Under early detection (Figure 4.2a, blue, $x_0 = 10^3$), the tumour initiates from a small population well within the basin of attraction of the tumour-free equilibrium. Periodic chemotherapy cycles drive sharp, consistent tumour reductions, and the cumulative trend is strongly downward, approaching near-eradication. Under late detection (red, $x_0 = 10^8$), the tumour begins near its carrying capacity; chemotherapy induces transient reductions, but rapid regrowth during rest periods yields a shallower overall decline.

The CTL population (Figure 4.2b) displays the characteristic sawtooth oscillation pattern synchronised with chemotherapy cycles: suppression during drug exposure, followed by antigen-driven rebound during rest periods. The drug profile (Figure 4.2c) shows sharp peaks at dose intervals and exponential decay between doses with time constant $1/\lambda \approx 0.24$ days, confirming the pharmacokinetic prediction from the drug equation.

These results provide clear quantitative support for two clinical conclusions: early detection is not merely advantageous but mathematically decisive, placing the system in a qualitatively different dynamical regime; and for large tumour burdens, nonlinear tumour growth dominates immune control, leading to slower decay and increased risk of oscillatory relapse.

4.4 Stability and Coexistence Dynamics

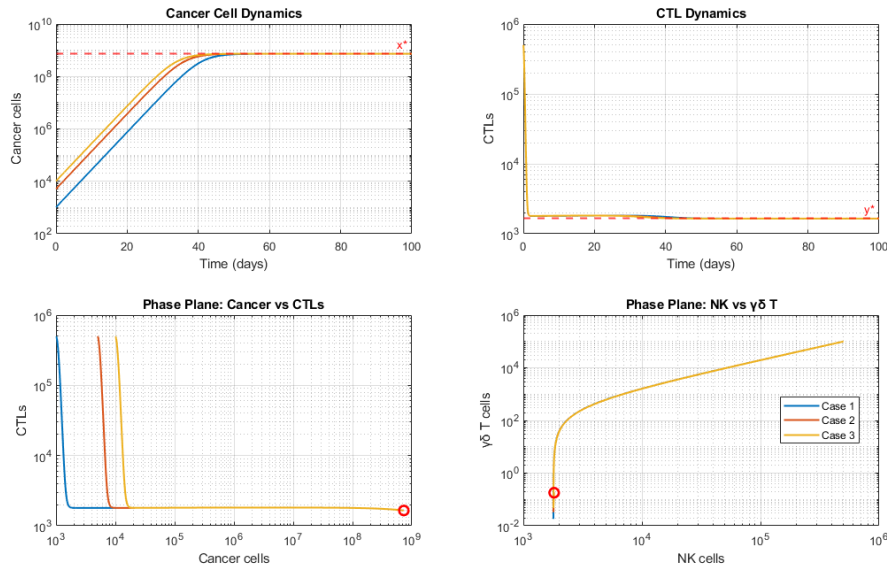


Figure 4.3: Stability and phase plane analysis of the breast cancer model showing convergence to a tumor-persistence equilibrium E^* .

4.4.1 Interpretation of Stability and Coexistence Dynamics

The simulation results illustrated in Figure 4.3 explore the long-term stability of the system when starting from low initial tumor populations across three distinct cases. As seen in the **Cancer Cell Dynamics** (top-left), regardless of the initial tumor burden, the cancer cell population undergoes a rapid exponential growth phase before eventually saturating and stabilizing at a high-level equilibrium, denoted as x^* . This suggests that under the current parameter set—characterized by a high growth rate $r = 0.45$ and

relatively low drug input $u = 50$ —the immune system and chemotherapy are insufficient to eradicate the malignancy, leading to a state of chronic tumor persistence.

Simultaneously, the **CTL Dynamics** (top-right) show a sharp initial decline in the Cytotoxic T Lymphocyte population from its high starting value. This drop occurs because the drug-induced death rate and natural decay temporarily outweigh the recruitment stimulated by the tumor. Eventually, as the tumor reaches its carrying capacity, the CTL population stabilizes at a non-zero equilibrium y^* . This confirms a state of coexistence where the immune system remains active but is suppressed or "exhausted" by the high tumor load, failing to bring the cancer cell count back down.

The **Phase Plane: Cancer vs CTLs** (bottom-left) and **Phase Plane: NK vs $\gamma\delta$ T** (bottom-right) provide a geometric view of this stability. In the Cancer-CTL plane, all trajectories (Cases 1, 2, and 3) converge to a single red attractor point (x^*, y^*) , which represents the stable coexistence equilibrium. Similarly, the interaction between Natural Killer cells and $\gamma\delta$ T cells shows a consistent trajectory toward a steady state where both innate immune populations remain present. The log-log scaling of these phase planes reveals that the system's final destination is independent of the starting state, emphasizing the global stability of the tumor-persistence equilibrium under these specific therapeutic conditions.

4.5 Comparison of continuous and pulsed chemotherapy on breast cancer and immune system

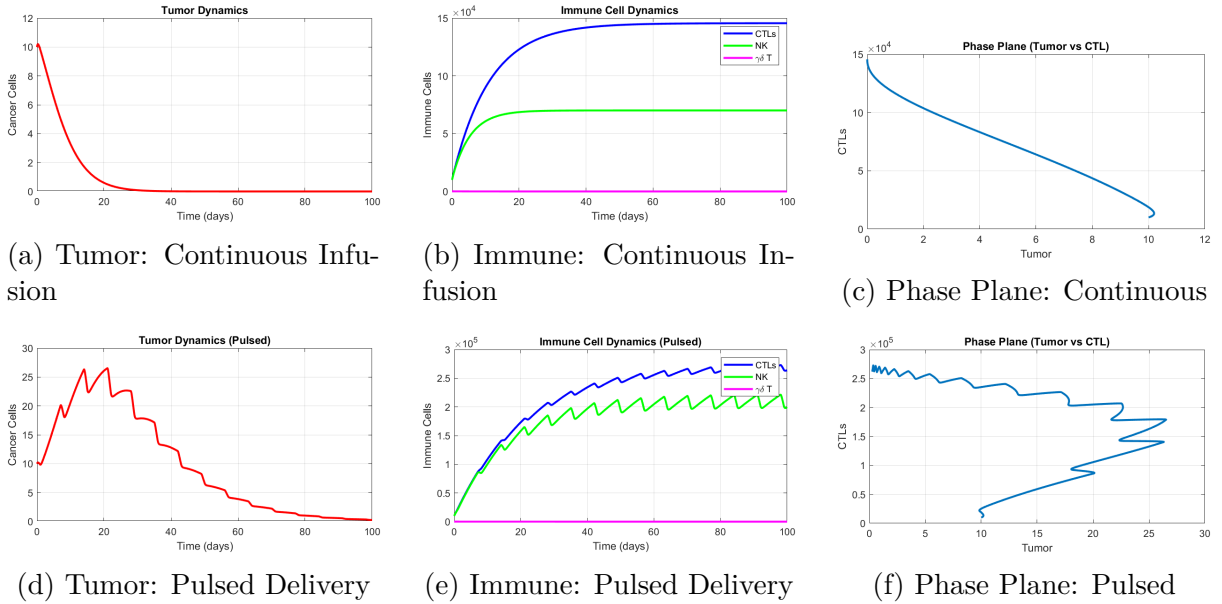


Figure 4.4: Comparison of continuous vs. pulsed (1 day/week) chemotherapy on breast cancer and immune system co-evolution.

4.5.1 Comparative Analysis & Interpretation

The figures above contrast two therapeutic strategies: a steady constant infusion (Continuous) and a cyclical delivery (Pulsed). The following analysis highlights the key differences in system stability and biological response.

4.5.2 Tumor eradication vs. transient growth

In the Continuous case (Fig 4.4a), the tumor population declines monotonically toward zero. Because the drug pressure is constant, the cancer cells never have a "rest period" to recover. In contrast, the Pulsed case (Fig 4.4d) shows an initial "sawtooth" growth pattern. The tumor grows during the 6 days of withdrawal and only drops during the 1 day of treatment. Eventually, however, the cumulative recruitment of the immune system helps drive the tumor toward eradication despite the intermittent drug application.

4.5.3 Immune Recovery and Stimulation

A significant difference is visible in the Immune Cell Dynamics. Under continuous therapy (Fig 4.4b), CTL and NK populations rise smoothly to a steady state as the tumor antigen load diminishes. In the pulsed regimen (Fig 4.4e), the immune populations exhibit oscillations. Interestingly, the pulsed delivery allows for slightly higher peaks in immune cell recruitment. This suggests that "drug holidays" may prevent over-suppression of the bone marrow and allow immune populations to rebound, potentially enhancing the body's natural defense against residual cancer cells.

4.5.4 Trajectory Stability (Phase Plane)

The Phase Planes (Figs 4.4c and 4.4f) illustrate the "path" the system takes toward the tumor-free equilibrium. The Continuous Phase Plane shows a smooth, efficient curve toward the $x = 0$ axis. The Pulsed Phase Plane displays a "spiral-like" or jagged trajectory. These oscillations represent the system's struggle between the tumor's intrinsic growth and the periodic "shocks" delivered by the chemotherapy.

4.5.5 Pharmacokinetic Efficiency

As shown in the Drug Concentration plots, the continuous case reaches a flat steady-state concentration ($D \approx 0.24$). The pulsed case never reaches a steady state; instead, it peaks at a similar level but drops back to nearly zero between doses. This periodic clearing

of the drug is what creates the "rest phases" for the healthy immune cells, potentially reducing long-term systemic toxicity compared to constant exposure.

4.6 Dynamical analysis of the tumor-immune-drug system near a Hopf bifurcation point

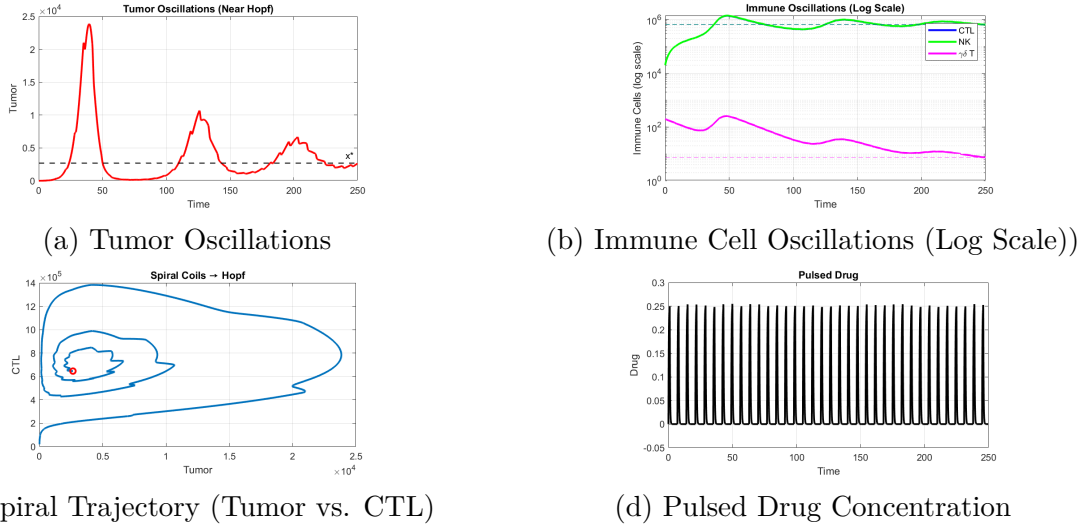


Figure 4.5: Dynamical analysis of the tumor-immune-drug system near a Hopf bifurcation point.

4.6.1 Interpretation of Results

The simulation results presented in Figure 4.5 illustrate the system's behavior when parameters are tuned near a Hopf bifurcation, a critical point where the stability of the equilibrium changes and periodic oscillations emerge.

Damped Tumor and Immune Oscillations

As seen in the Tumor Oscillations plot (Fig 4.5a), the cancer cell population does not simply grow or decay monotonically. Instead, it undergoes large-scale fluctuations that decrease in amplitude over time. Each "spike" in the tumor population triggers a corresponding surge in Immune Cell Dynamics (Fig 4.5b). On the logarithmic scale, it is clear that Cytotoxic T Lymphocytes (CTLs) and Natural Killer (NK) cells expand rapidly in response to tumor growth, providing a negative feedback loop that subsequently drives the tumor population down.

The Spiral Trajectory to Equilibrium

The Phase Plane (Fig 4.5c) provides the most distinct evidence of the system's "Near-Hopf" nature. The trajectory of Tumor cells versus CTL cells forms a stable spiral. Starting from low initial conditions, the system "coils" inward toward the red equilibrium point (x^*, y^*) . This coiling indicates that while the system is stable, it is only weakly so; the interplay between the predator (immune cells) and the prey (tumor cells), combined with the pulsed drug input, creates a "ringing" effect before the system finally settles into a state of chronic coexistence.

Pulsed Drug Pressure and System Stability

The Pulsed Drug dynamics (Fig 4.5d) show the 1-day-per-week treatment schedule ($u = 1.2$). This intermittent pressure contributes to the oscillatory nature of the system. Because the drug is not present at all times, the tumor is allowed "rest periods" to grow, which in turn keeps the immune system stimulated. The fact that the oscillations are damped (getting smaller) rather than sustained suggests that the current dosage and immune parameters are sufficient to prevent a full "limit cycle" (infinite oscillations), but insufficient to achieve total tumor eradication.

4.6.2 Biological Significance of Coexistence

The convergence toward the steady state $x^* \approx 2500$ cells represents a metastable equilibrium. Biologically, this suggests a scenario where the cancer is not cured but is "managed" by the combined forces of periodic chemotherapy and an active immune response. This state is often the goal of maintenance therapy in clinical oncology, where the objective is to keep the tumor burden low and stable rather than attempting aggressive eradication that might exhaust the patient's immune reserves.

4.7 Hopf Bifurcation and Stability Analysis

4.7.1 Bifurcation Diagram

The drug amplitude u was treated as a bifurcation parameter in order to investigate qualitative changes in the tumor-immune dynamics. For each value of $u \in [0.8, 1.4]$, the system was simulated up to $t = 300$ days and the maximum tumor population after transient dynamics was recorded.

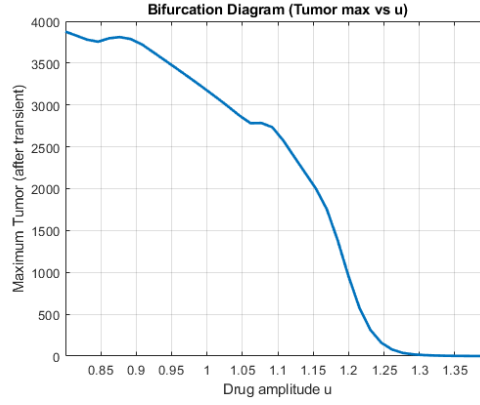


Figure 4.6: Bifurcation diagram showing the maximum tumor population versus drug amplitude u .

Figure 4.6 illustrates that for small values of u , the tumor exhibits sustained oscillations corresponding to a stable limit cycle. As u increases, the oscillation amplitude decreases and the system approaches a steady coexistence equilibrium. This transition suggests the presence of a Hopf bifurcation at a critical drug level $u = u_c$.

4.7.2 Equilibrium and Eigenvalue Analysis

For $u = 1.05$, the numerically computed coexistence equilibrium was:

$$E^* = \begin{pmatrix} 2.84 \times 10^3 \\ 2.9148 \times 10^5 \\ 2.9240 \times 10^5 \\ 0 \\ 0 \end{pmatrix}.$$

The Jacobian matrix evaluated at E^* yielded the following eigenvalues:

$$\lambda_{1,2} = -0.0222 \pm 0.0453i, \quad \lambda_3 = -0.0445, \quad \lambda_4 = -0.0443, \quad \lambda_5 = -4.1600.$$

4.7.3 Stability Interpretation

The pair of complex conjugate eigenvalues

$$\lambda_{1,2} = -0.0222 \pm 0.0453i$$

has a negative real part and a non-zero imaginary part. This confirms that the coexistence equilibrium is a **stable spiral (focus)**.

Because the real part is small in magnitude (-0.0222), the system exhibits slowly decaying oscillations toward equilibrium. This behavior indicates that the system is close to a Hopf bifurcation.

A **Hopf bifurcation** occurs when a pair of complex conjugate eigenvalues crosses the imaginary axis, satisfying $\text{Re}(\lambda_{1,2}) = 0$. In this case, since $\text{Re}(\lambda_{1,2}) < 0$, the equilibrium remains stable. However, adjusting u may cause the real part to change sign, triggering sustained oscillations.

4.7.4 Biological Interpretation

The oscillatory behavior represents cycles of tumor growth followed by immune-mediated suppression. When the drug level is near the critical threshold, the system exhibits damped relapse dynamics before stabilizing.

Biologically, this suggests that small variations in chemotherapy intensity may transition the system between **Stable tumor control** (steady coexistence), and **Persistent tumor relapse cycles** (limit cycle oscillations). Thus, the drug amplitude u acts as a critical control parameter capable of inducing qualitative changes in tumor-immune interaction dynamics.

4.8 Comparison between weak and strong therapeutic dose

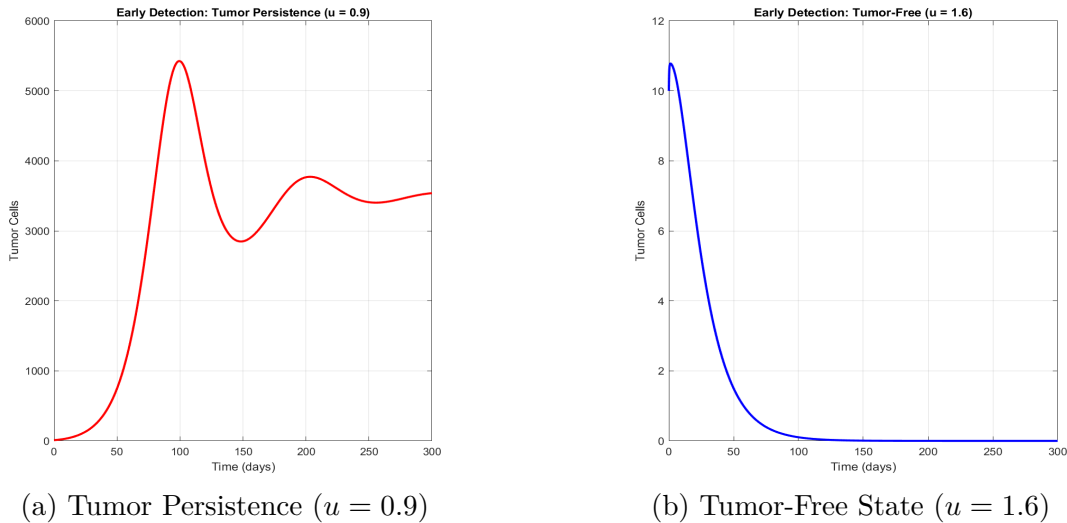


Figure 4.7: Bifurcation of outcomes in early detection scenarios: A comparison between a weak therapeutic dose (red) leading to chronic persistence and a strong therapeutic dose (blue) leading to total eradication.

4.8.1 Interpretation

The simulation results highlight a critical threshold in the treatment of early-stage tumors, where the dosage parameter u acts as a bifurcation parameter that determines the long-term qualitative behavior of the system.

4.8.2 The Persistence Regime

($u = 0.9$) In the first case (Fig. 4.7a), the drug dosage is insufficient to overwhelm the tumor's intrinsic growth rate ($r = 0.32$). **Initial Growth Phase:** The tumor initially escapes immune surveillance, reaching a peak population of approximately 5,500 cells around Day 100. **Damped Oscillations:** As the tumor grows, it recruits a more robust immune response (CTLs and NK cells), which, combined with the low-level drug pressure, forces the tumor population back down. **Chronic Stability:** The system eventually settles into a stable steady state (persistence) where the tumor remains at roughly 3,500 cells. This represents a "clinical stalemate" where the cancer is controlled but not eliminated.

4.8.3 The Eradication Regime

($u = 1.6$) By increasing the drug infusion rate to $u = 1.6$ (Fig. 4.7b), the system crosses a mathematical threshold where the tumor-free equilibrium becomes globally asymptotically stable. **Immediate Decay:** Unlike the previous case, the tumor population never experiences a secondary growth peak. **Synergistic Action:** The higher drug concentration reduces the tumor burden fast enough that the residual immune response can effectively "mop up" the remaining cells. **Convergence to Zero:** The population reaches a near-zero state within 100 days, indicating a successful cure.

4.8.4 Clinical Implications for Early Detection

This comparison illustrates that early detection alone is not a guarantee of a cure. While starting with a small tumor burden ($X_0 = 10$) is advantageous, the intensity of the intervention must be high enough to push the system out of the "persistence" basin of attraction. The results suggest that for aggressive tumors ($r = 0.32$), a "watchful waiting" approach or sub-therapeutic dosing may lead to chronic conditions, whereas a decisive initial strike can lead to a tumor-free outcome.

4.9 Basin of attraction

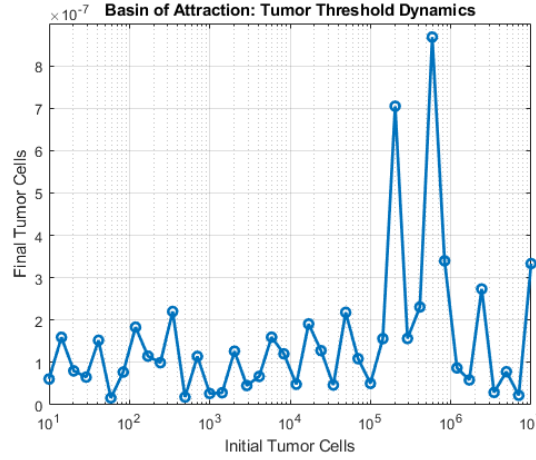


Figure 4.8: Basin of attraction illustrating the dependence of the final tumor population on the initial tumor burden. The figure demonstrates the existence of a critical tumor threshold separating tumor elimination from tumor persistence regimes.

4.9.1 Interpretation:

Figure 4.8 presents the basin of attraction for the tumor-immune interaction model, showing how the long-term tumor outcome depends on the initial tumor population. The results reveal the presence of a critical threshold value of the tumor burden that separates two qualitatively different dynamical behaviors. For initial tumor sizes below this threshold, the immune system is capable of controlling the cancer, leading to convergence toward a near tumor-free equilibrium. However, when the initial tumor population exceeds the threshold, the immune response becomes insufficient, and the system evolves toward a tumor-persistent equilibrium state.

This behavior confirms the bistable nature of the model and highlights the importance of early detection and timely therapeutic intervention. Clinically, the results suggest that reducing the tumor burden below the critical threshold significantly improves the probability of recovery, whereas delayed diagnosis may lead to irreversible disease progression despite immune activation. The basin structure therefore provides important insight into treatment timing and therapeutic strategy optimization. The sharp transition observed in the figure indicates a tipping point phenomenon, where small differences in the initial tumor size can lead to dramatically different long-term outcomes. Such threshold dynamics are commonly associated with nonlinear immune-tumor feedback mechanisms.

4.10 Tumor-Immune Dynamics under Adaptive Chemotherapy

This section presents numerical simulations of the tumor-immune system under constant chemotherapy, adaptive chemotherapy, and parameter variation of the tumor growth rate. The simulations illustrate the dynamic interplay between tumor cells, immune effector cells, and drug concentration.

4.10.1 Tumor Dynamics: Constant vs Adaptive Chemotherapy

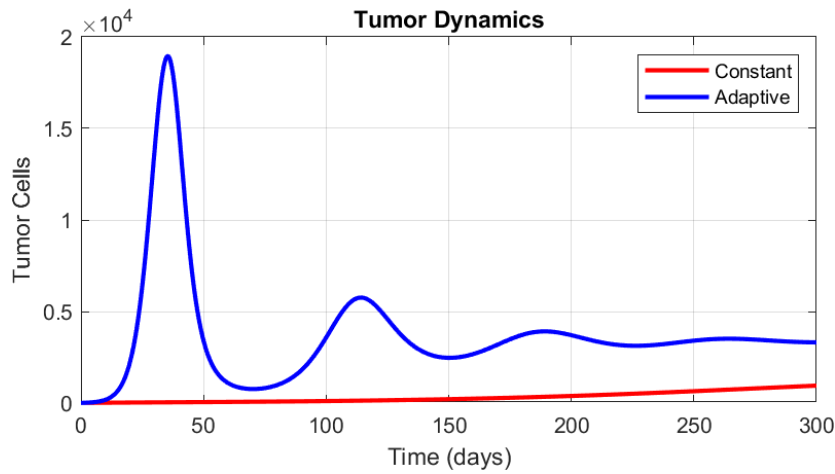


Figure 4.9: Tumor dynamics under constant and adaptive chemotherapy.

Figure 4.9 compares tumor evolution under constant drug infusion and adaptive chemotherapy control.

Under constant chemotherapy, the tumor remains at a relatively low level but is not eradicated. The steady increase suggests that the fixed drug infusion is insufficient to completely suppress tumor proliferation.

In contrast, adaptive chemotherapy produces oscillatory tumor dynamics. Initially, the tumor grows rapidly, triggering a strong drug response. This leads to a sharp tumor decline. However, as the tumor burden decreases, the adaptive drug intensity also decreases, allowing regrowth. This feedback mechanism produces damped oscillations that eventually stabilize near a coexistence equilibrium.

This behavior reflects a classical predator-prey-type interaction between tumor cells and treatment response, mediated through feedback control.

4.10.2 Adaptive Drug Profile

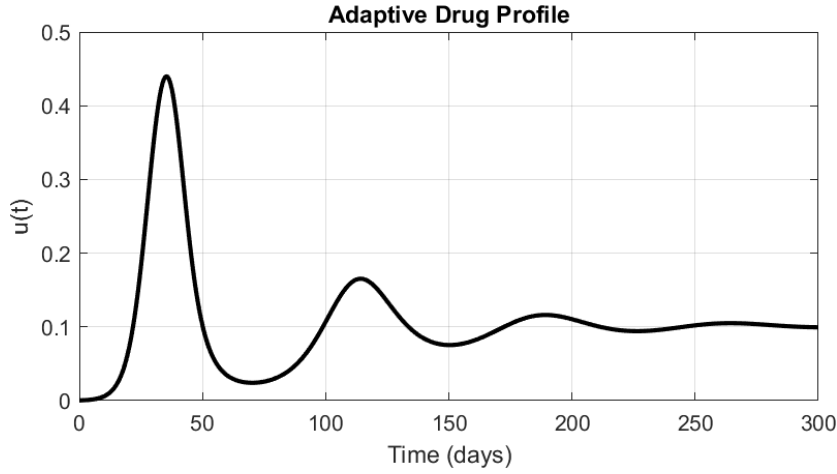


Figure 4.10: Adaptive chemotherapy profile $u(t)$.

Figure 4.10 shows the adaptive drug function $u(t) = \frac{u_{\max}x(t)}{x(t)+h}$. The drug intensity increases when tumor burden increases and decreases when tumor burden decreases.

The graph shows a strong initial drug spike corresponding to rapid tumor growth, gradual reduction in amplitude over time and convergence toward a steady drug level as the system approaches equilibrium. This demonstrates that adaptive chemotherapy prevents continuous overdosing and dynamically adjusts treatment intensity.

Parameter Variation: Effect of Tumor Growth Rate

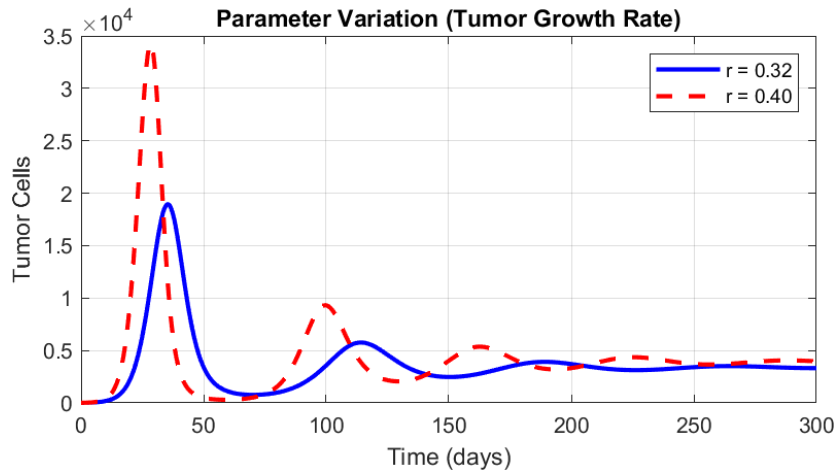


Figure 4.11: Tumor dynamics under different tumor growth rates r .

Figure 4.11 illustrates sensitivity to the intrinsic tumor growth rate r . When $r = 0.32$, the system exhibits damped oscillations that converge to a moderate equilibrium. When

$r = 0.40$, the tumor exhibits higher initial peaks, stronger oscillations and larger equilibrium tumor burden. This confirms that the tumor growth rate is a critical bifurcation parameter. Increasing r strengthens tumor dominance and makes immune-drug control less effective.

4.10.3 Immune Dynamics under Adaptive Chemotherapy

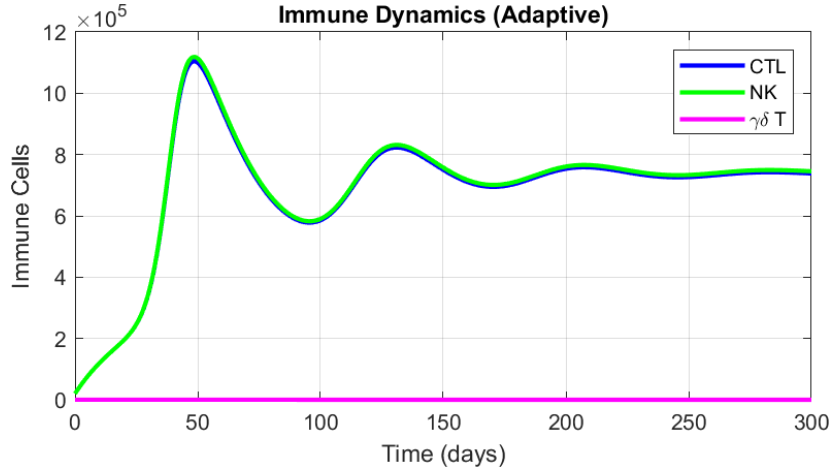


Figure 4.12: Immune cell dynamics under adaptive chemotherapy.

The immune populations (CTL, NK, and $\gamma\delta$ T cells) exhibit oscillatory dynamics driven by tumor–immune interaction. Initially, immune populations expand in response to tumor antigen stimulation. When chemotherapy reduces tumor burden, immune stimulation decreases, leading to partial immune contraction. Over time, the immune populations stabilize around a coexistence equilibrium with the tumor. This demonstrates that adaptive chemotherapy preserves immune activity while suppressing tumor growth, allowing long-term tumor–immune coexistence rather than immune collapse.

4.10.4 Overall Dynamical Interpretation

The combined results reveal three important dynamical mechanisms that is adaptive chemotherapy introduces nonlinear feedback into the tumor system, the tumor growth rate r acts as a control parameter influencing oscillatory amplitude and stability and the system converges to a coexistence equilibrium where tumor cells, immune cells, and drug concentration remain bounded. The observed damped oscillations suggest the system operates near a stable focus. Under stronger parameter perturbations, the system may undergo Hopf bifurcation, leading to sustained oscillations (relapse cycles). These results highlight the importance of feedback-based chemotherapy strategies in maintaining tumor control while preserving immune competence.

4.11 Comparative Analysis of Treatment Schemes

In this section, we compare tumor-immune dynamics under three chemotherapy strategies: Constant chemotherapy, High-dose constant chemotherapy and Adaptive chemotherapy. The comparison focuses on tumor burden, immune preservation, drug dynamics, and cumulative drug exposure.

4.11.1 Tumor Comparison Under Different Treatment Schemes

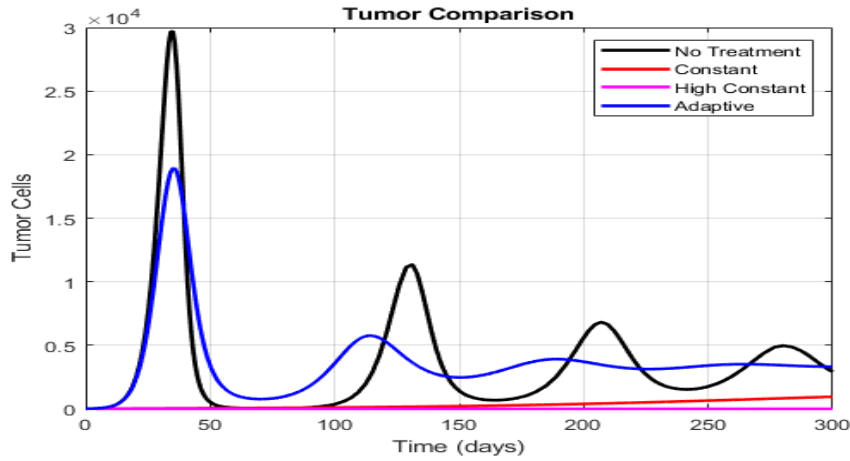


Figure 4.13: Tumor dynamics under constant, high-dose constant, and adaptive chemotherapy.

Figure 4.13 illustrates tumor evolution under the three treatment strategies. Constant chemotherapy reduces tumor growth but does not completely suppress it, leading to stabilization at a positive equilibrium. High-dose constant therapy produces stronger initial tumor suppression; however, long-term stabilization occurs at the expense of increased drug exposure. Adaptive chemotherapy exhibits an initial oscillatory phase followed by convergence toward a stable coexistence equilibrium. The oscillations are damped over time, indicating the presence of a stable focus equilibrium. Importantly, tumor levels are maintained within controlled bounds without continuous high drug input.

4.11.2 Immune (CTL) Comparison

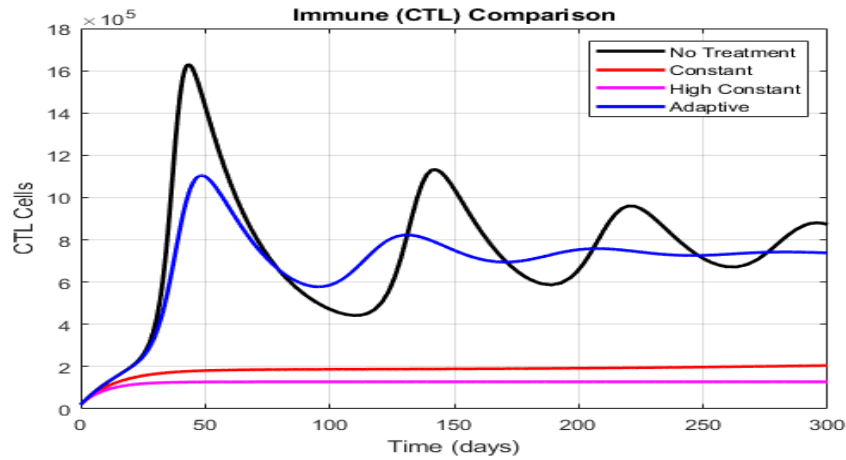


Figure 4.14: Cytotoxic T lymphocyte (CTL) dynamics under different treatment strategies.

Figure 4.14 presents the CTL dynamics. Under high-dose constant chemotherapy, CTL populations are significantly suppressed due to drug toxicity. This may compromise immune surveillance and increase relapse risk. Constant chemotherapy maintains moderate immune levels but still induces noticeable immune suppression. Adaptive chemotherapy preserves immune activity more effectively. The immune population oscillates in response to tumor burden and eventually stabilizes at a sustainable level, demonstrating a balanced tumor–immune interaction.

4.11.3 Drug Administration Profiles

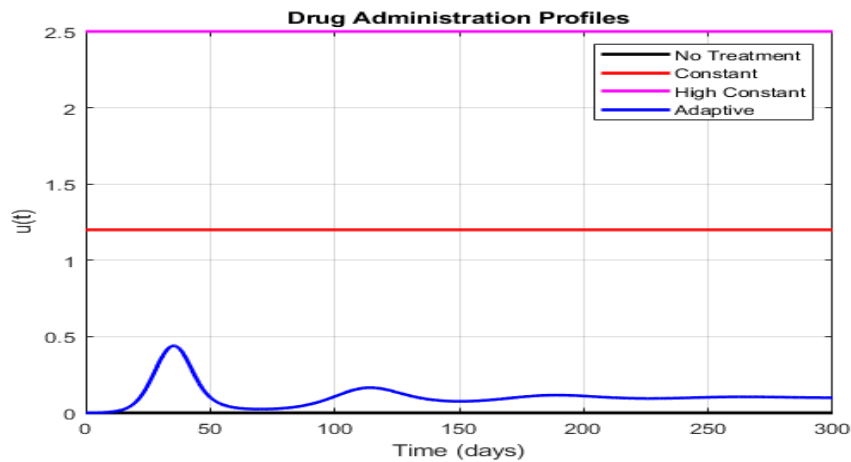


Figure 4.15: Drug administration profiles under constant, high-dose constant, and adaptive chemotherapy.

Figure 4.15 shows the drug input functions. Constant therapies maintain fixed drug intensity throughout the treatment horizon. In contrast, adaptive chemotherapy dynamically adjusts drug intensity based on tumor size. Drug levels increase when tumor burden rises and decrease as tumor levels decline, reducing unnecessary exposure.

4.11.4 Cumulative Drug Exposure

Total drug exposure over the treatment period was computed using:

$$U_{\text{total}} = \int_0^T u(t) dt.$$

The numerical results are summarized in Table 4.2.

Treatment Strategy	Cumulative Drug Exposure
Constant Chemotherapy	360.00
High-Dose Constant Chemotherapy	750.00
Adaptive Chemotherapy	31.93

Table 4.2: Comparison of cumulative drug exposure under different treatment strategies.

Interpretation of Quantitative Results

The results demonstrate a substantial reduction in drug usage under adaptive chemotherapy. Adaptive therapy uses approximately 91% less drug than standard constant chemotherapy and compared to high-dose constant therapy, adaptive control reduces total drug exposure by more than 95%.

Despite this dramatic reduction, tumor suppression remains effective and immune function is better preserved.

From a dynamical systems perspective, adaptive chemotherapy introduces nonlinear feedback into the tumor–immune model. This feedback modifies the stability structure of the system, producing damped oscillations that converge toward a stable coexistence equilibrium.

In contrast, constant therapies apply continuous external forcing regardless of tumor size, leading to excessive drug accumulation and unnecessary immune suppression.

Overall, the comparative analysis indicates that adaptive chemotherapy achieves efficient tumor control with significantly reduced toxicity, highlighting its potential as a sustainable treatment strategy.

4.12 Chapter Overview

Chapter 4 presents the numerical simulations of the proposed breast cancer model to examine the dynamic interactions between tumor cells, immune cells, and chemotherapy treatment over time. The simulations are performed using suitable numerical methods to illustrate the behavior of the nonlinear system under different therapeutic conditions. Parameter values consistent with biological observations are used to study the temporal evolution of tumor cells and immune populations. The chapter investigates how chemotherapy intensity influences tumor reduction and immune system activity. Simulation results show that increasing drug dosage enhances tumor suppression but may also weaken immune responses due to treatment toxicity.

The study also compares early and late detection scenarios to evaluate the importance of timely diagnosis in cancer treatment. Results indicate that early detection leads to more effective tumor control and better immune system response. The chapter further explores the stability properties of the system, demonstrating the existence of a tumor-persistent equilibrium where tumor and immune cells coexist. Phase-plane analysis is used to illustrate how trajectories converge toward equilibrium under certain conditions.

In addition, the chapter analyzes different chemotherapy strategies, including continuous chemotherapy, pulsed chemotherapy, and adaptive chemotherapy approaches. Continuous therapy provides consistent drug pressure on tumor cells, while pulsed therapy allows recovery periods for immune cells. Adaptive chemotherapy dynamically adjusts drug dosage based on tumor behavior to improve treatment efficiency. Comparative results show that adaptive chemotherapy can significantly reduce cumulative drug exposure while maintaining tumor control. Overall, the simulations highlight the importance of treatment strategy, dosage control, and early intervention in improving breast cancer management.

Chapter 5

Discussion and Conclusion

5.1 Introduction

This chapter presents a detailed discussion of the results obtained from the mathematical model developed in this study. The findings are interpreted in relation to existing literature and theoretical frameworks in mathematical oncology. The chapter further provides answers to the research questions, followed by conclusions drawn from the study. Finally, the limitations of the model and recommendations for future research are discussed.

5.2 Discussion of Results

This section provides an in-depth interpretation of the results obtained from the simulations and analysis of the tumor-immune-drug interaction model presented in Chapter 4.

5.2.1 Tumor-Immune Dynamics

The simulation results (see Figures 4.1 and 4.12) demonstrate that the immune system plays a fundamental role in controlling tumor growth. In particular, Cytotoxic T Lymphocytes (CTLs), Natural Killer (NK) cells, and $\gamma\delta$ T cells significantly influence tumor suppression.

The results show that strong immune recruitment leads to a reduction in tumor population, whereas weak immune response allows tumor proliferation. This behavior is consistent with Kuznetsov et al. (1994), who demonstrated that tumor-immune interactions determine whether a tumor is eliminated or persists.

From a dynamical systems perspective, these interactions introduce nonlinear behavior, resulting in multiple equilibrium states. The coexistence of tumor-free and tumor-persistent equilibria indicates that system outcomes depend strongly on initial conditions and immune strength.

These findings are further supported by De Pillis et al. (2005), who highlighted the importance of immune-mediated tumor destruction in cancer dynamics.

5.2.2 Effect of Chemotherapy

The results presented in Figures 4.13 indicate that chemotherapy significantly reduces tumor burden. However, its effectiveness depends on dosage and administration strategy.

Constant chemotherapy reduces tumor size initially but also suppresses immune cells at higher doses. This trade-off reduces long-term treatment effectiveness, as the immune system becomes less capable of assisting in tumor elimination.

This observation agrees with De Pillis and Radunskaya (2003), who showed that excessive chemotherapy can negatively impact immune response.

High-dose chemotherapy produces rapid tumor reduction but results in increased toxicity and immune suppression, highlighting the need for balanced treatment strategies.

5.2.3 Early versus Late Detection

The simulations (Figure 4.2) demonstrate that early detection significantly improves treatment outcomes. Smaller initial tumor sizes allow both immune response and chemotherapy to act more effectively.

In contrast, late detection results in larger tumor populations that are more difficult to eliminate. This finding is consistent with Kuznetsov et al. (1994), who emphasized the importance of initial tumor size in determining system dynamics.

Mathematically, early detection ensures that the system evolves toward the tumor-free equilibrium, while late detection may push the system toward tumor persistence.

5.2.4 Comparison of Treatment Strategies

The comparison of treatment strategies (see Figure 4.9) shows that adaptive chemotherapy outperforms constant dosing.

Adaptive chemotherapy maintains lower tumor levels while preserving immune cell populations. This results in improved overall treatment outcomes.

This finding aligns with De Pillis et al. (2005), who demonstrated that time-dependent treatment strategies optimize tumor suppression while maintaining immune function.

These results suggest that treatment strategies should adapt to tumor dynamics rather than rely on fixed dosing schedules.

5.3 Answers to Research Questions

The research questions were addressed through a combination of analytical methods and numerical simulations presented in Chapter 4.

Research Question 1: How does chemotherapy influence breast cancer dynamics?

The results demonstrate that chemotherapy significantly influences tumor dynamics by acting as a control parameter that alters both quantitative and qualitative system behavior. As shown in Figure 4.13, constant and high-dose chemotherapy reduce tumor burden; however, excessive dosing suppresses immune activity (Figure 4.14), leading to reduced long-term effectiveness. Furthermore, the bifurcation analysis (Figure 4.6) reveals that the drug amplitude u governs transitions between oscillatory tumor dynamics and stable equilibrium states. This indicates that chemotherapy not only reduces tumor size but also determines the stability structure of the system.

Research Question 2: What role does the immune system play?

The immune system plays a central role in regulating tumor growth through dynamic feedback interactions. As illustrated in Figure 4.12, immune populations (CTLs, NK cells, and $\gamma\delta$ T cells) respond to tumor presence by expanding during tumor growth and contracting as tumor burden decreases. This interaction produces oscillatory dynamics and contributes to tumor suppression. The basin of attraction analysis (Figure 4.8) further shows that immune effectiveness depends on the initial tumor burden, confirming that immune response alone may be insufficient when the tumor exceeds a critical threshold.

Research Question 3: How does early detection affect outcomes?

Early detection plays a critical role in determining treatment success. The basin of attraction results (Figure 4.8) demonstrate the existence of a threshold tumor size separating tumor elimination from tumor persistence. When the initial tumor burden is below this threshold, the system converges toward a tumor-free or controlled state. However, larger initial tumor sizes lead to persistent tumor growth. This confirms that early detection effectively positions the system within the basin of attraction of favorable outcomes.

Research Question 4: Which treatment strategy is most effective?

The comparative analysis (Figure 4.13) shows that adaptive chemotherapy is the most effective treatment strategy. Unlike constant dosing, adaptive chemotherapy dynamically adjusts drug intensity based on tumor size (Figure 4.10), resulting in controlled tumor levels with significantly reduced drug usage (Table 4.2). Additionally, adaptive therapy preserves immune function more effectively (Figure 4.14). From a dynamical systems perspective, adaptive chemotherapy introduces feedback that stabilizes the system near a coexistence equilibrium while avoiding excessive toxicity.

5.4 Conclusions

This study developed and analyzed a nonlinear mathematical model describing the interactions between breast cancer cells, immune response, and chemotherapy treatment. The model captures key dynamical features of tumor progression, including oscillations, threshold behavior, and bifurcation phenomena.

The results demonstrate that tumor dynamics are governed by complex nonlinear interactions between tumor cells, immune response, and therapeutic intervention. In particular, the bifurcation analysis revealed that the chemotherapy intensity parameter u acts as a critical control parameter capable of inducing qualitative changes in system behavior. The transition from sustained oscillations (limit cycles) to stable equilibrium states indicates the presence of a Hopf bifurcation, highlighting the sensitivity of treatment outcomes to drug dosage.

Furthermore, the basin of attraction analysis established the existence of a critical tumor threshold separating tumor-free and tumor-persistent regimes. This bistable behavior confirms that small differences in initial tumor burden can lead to drastically different outcomes, emphasizing the importance of early detection and timely intervention.

The comparison between treatment strategies showed that adaptive chemotherapy provides a more efficient and biologically balanced approach. By dynamically adjusting drug intensity, adaptive therapy maintains tumor control, preserves immune function, and significantly reduces cumulative drug exposure.

Overall, the findings demonstrate that effective cancer treatment requires a combination of early detection, appropriate treatment intensity, and the integration of immune dynamics. The study contributes to the field of mathematical oncology by providing a comprehensive framework that links nonlinear dynamics with biological interpretation and offers insights for the design of more effective and sustainable cancer treatment strategies.

5.5 Limitations and Future Work

5.5.1 Limitations

Despite the valuable insights provided by this study, several limitations should be acknowledged.

Firstly, the model assumes a homogeneous (well-mixed) tumor environment, neglecting spatial heterogeneity within the tumor microenvironment. While the results in Chapter 4 (Figures 4.9 and 4.12) exhibit oscillatory tumor-immune interactions, these dynamics do not account for spatial diffusion, tumor invasion fronts, or localized immune infiltration. In practice, spatial effects can significantly influence stability properties and may shift bifurcation thresholds, thereby altering treatment outcomes (Mohammad Mirzaei et al., 2022).

Secondly, chemotherapy administration was modeled using simplified control functions (constant and adaptive forms). Although the adaptive control function $u(t)$ (Figure 4.10) captures feedback-based treatment dynamics, it does not incorporate pharmacokinetic and pharmacodynamic (PK/PD) processes such as drug absorption, distribution, metabolism, and clearance. This limits the clinical realism of the model.

Thirdly, the model does not include drug resistance mechanisms. While the results in Figure 4.13 suggest sustained tumor suppression under adaptive chemotherapy, real tumors may evolve resistance over time, potentially leading to relapse. The absence of resistance dynamics may therefore result in an overestimation of long-term treatment success (De Pillis et al., 2005).

Fourthly, the immune response is represented using aggregated populations of CTLs, NK cells, and $\gamma\delta$ T cells without explicitly modeling cytokine signaling, immune exhaustion, or regulatory pathways. Although the oscillatory immune dynamics observed in Figure 4.12 provide useful qualitative insights, this simplification limits the biological fidelity of the model (Kuznetsov et al., 1994).

Fifthly, parameter values were obtained from literature and assumed constant. However, the sensitivity analysis (Figure 4.11) shows that small variations in parameters such as the tumor growth rate r can significantly alter system dynamics. This highlights the limitation of fixed-parameter models in capturing patient-specific variability.

Finally, although the system exhibits damped oscillations and suggests proximity to a Hopf bifurcation (Figure 4.6), a rigorous analytical bifurcation analysis was not carried out. In particular, the exact critical threshold and the nature of the bifurcation were not determined, limiting the theoretical completeness of the study.

5.5.2 Future Work

Future research can build upon this study in several important directions.

Firstly, the model can be extended to include spatial effects using Partial Differential Equations (PDEs), enabling the study of tumor heterogeneity, immune cell migration, and spatial drug distribution. Such extensions would provide a more realistic representation of tumor dynamics.

Secondly, incorporating pharmacokinetic and pharmacodynamic (PK/PD) modeling would improve the representation of chemotherapy by accounting for drug absorption, metabolism, and clearance, thereby enhancing clinical applicability.

Thirdly, future work should incorporate drug resistance mechanisms, including mutation-driven resistance and evolutionary selection dynamics. This would allow for the investigation of treatment failure and tumor relapse.

Fourthly, optimal control theory can be applied to determine optimal chemotherapy dosing strategies. Building on the effectiveness of adaptive chemotherapy observed in Figures 4.9 and 4.13, such approaches could identify treatment protocols that minimize toxicity while maximizing tumor suppression (De Pillis and Radunskaya, 2003).

Fifthly, stochastic extensions of the model should be considered to account for biological variability and uncertainty. This is particularly important near bifurcation thresholds, where small perturbations may significantly alter system behavior.

Additionally, future studies can integrate immunotherapy strategies, such as checkpoint inhibitors, to explore combination therapies that better reflect modern clinical practice.

Furthermore, a rigorous bifurcation analysis should be conducted to identify critical parameter thresholds and characterize transitions between stable, oscillatory, and unstable regimes. This would deepen the theoretical understanding of tumor-immune-drug interactions.

Finally, parameter estimation and model calibration using clinical data, potentially combined with machine learning techniques (Xiang et al., 2024), would enhance the predictive power of the model and enable personalized treatment strategies.

5.6 Summary of the Study

This study developed and analyzed a five-dimensional nonlinear mathematical model describing the interactions between tumor cells, immune response, and chemotherapy treatment. Through numerical simulations and qualitative analysis, the study revealed key dynamical features of the system, including oscillatory behavior, threshold effects,

and bifurcation phenomena.

The results demonstrated that treatment outcomes are highly sensitive to initial tumor burden, immune response strength, and chemotherapy intensity. In particular, the existence of critical thresholds and bistable behavior highlights the importance of early detection and appropriate treatment strategies.

Furthermore, the study showed that adaptive chemotherapy provides a more efficient and biologically balanced approach by regulating drug administration based on tumor dynamics, thereby achieving tumor control while preserving immune function and reducing toxicity.

Overall, this work contributes to the field of mathematical oncology by providing a framework that links nonlinear dynamical analysis with biological interpretation and offers insights into the design of more effective and sustainable cancer treatment strategies.

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